

CHAPTER 7

Effect of the shear force in bending

CONTENT

1 – Introduction

2 – Shear stress

3 – Shear reinforcement verifications

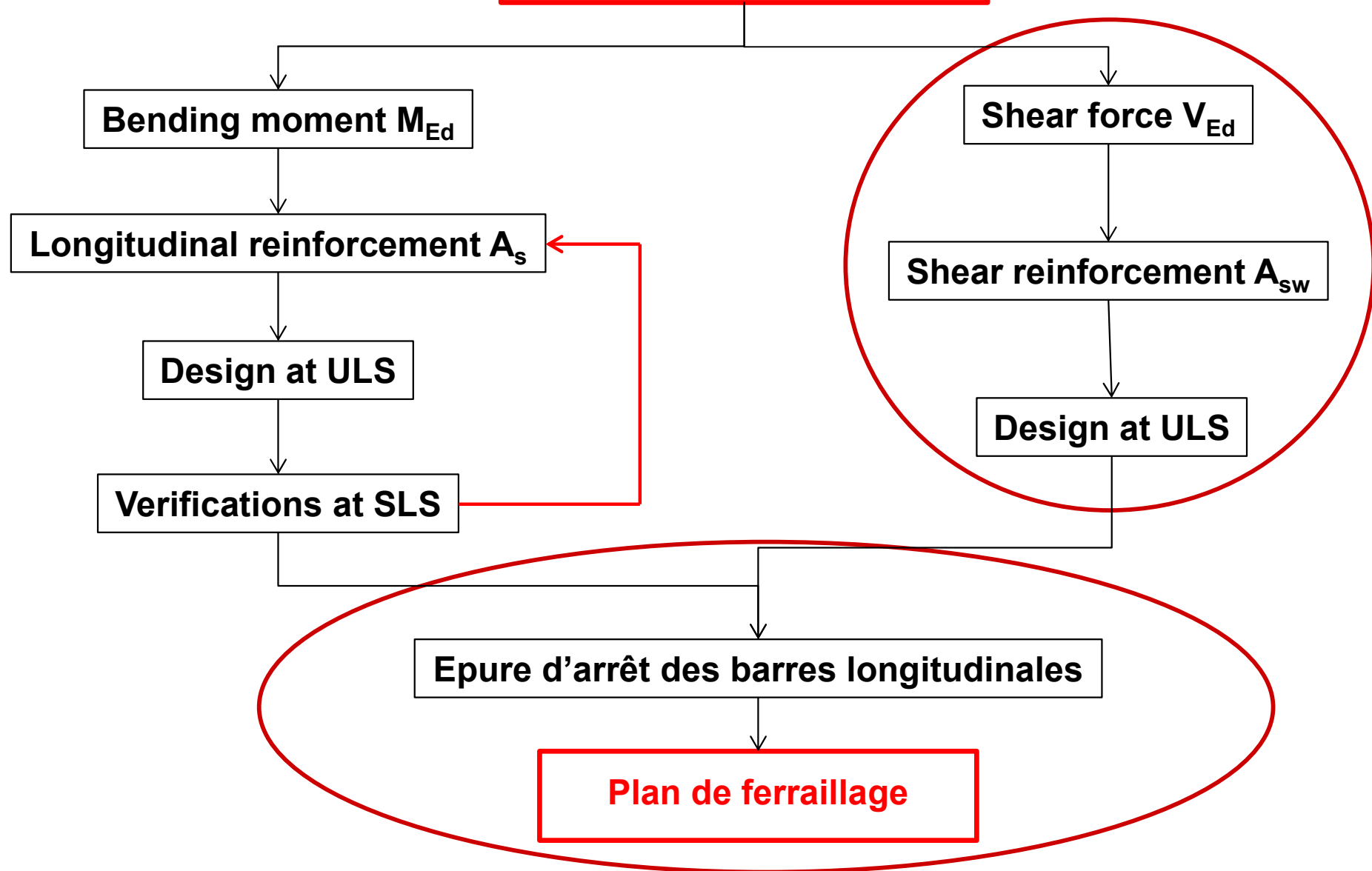
4 – Curtailment of longitudinal reinforcement

5 – Verification at supports

6 – Flange and web connection

1 – Introduction

Bending without axial load



2 – Shear stress

Reminder: Calcul des contraintes tangentes

Réciprocité des contraintes de Cauchy

$$\tau_{yx} = \tau_{xy} = \tau$$

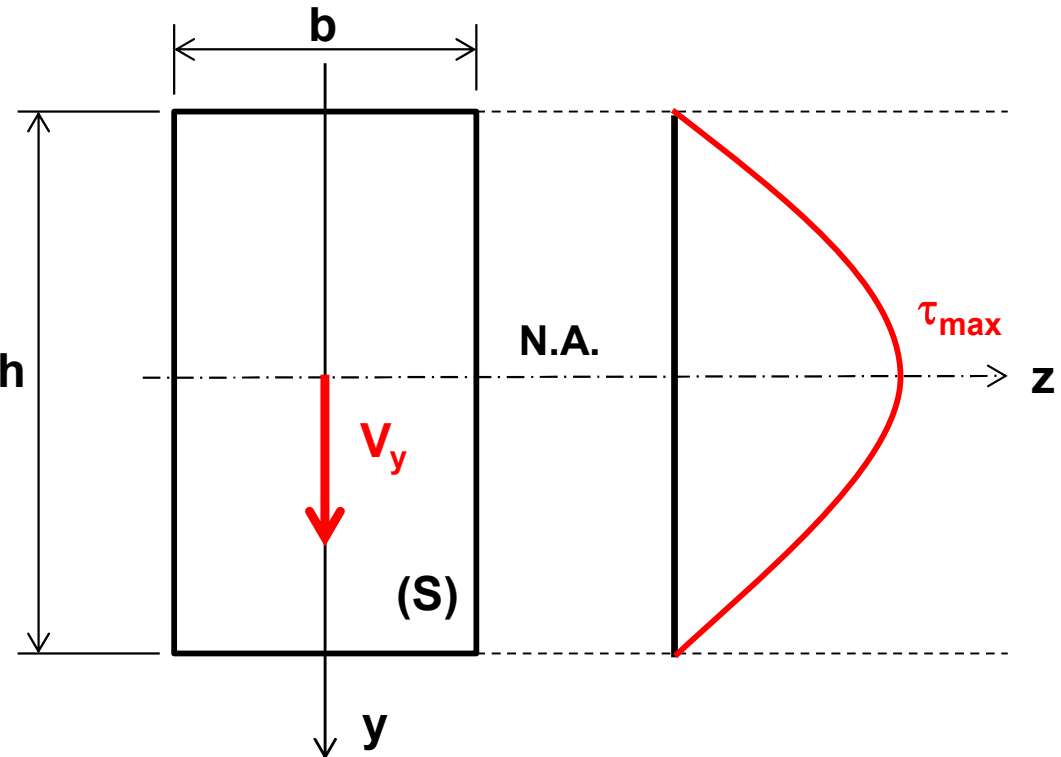
Shear stress

$$\tau(y) = \frac{V_y}{b(y)} \frac{\mu_z(y)}{I}$$

Where $\mu_z(y)$ = 1st moment of area from z

$$\mu_z(h/2) = \mu_z(-h/2) = 0$$

I = 2nd moment of area

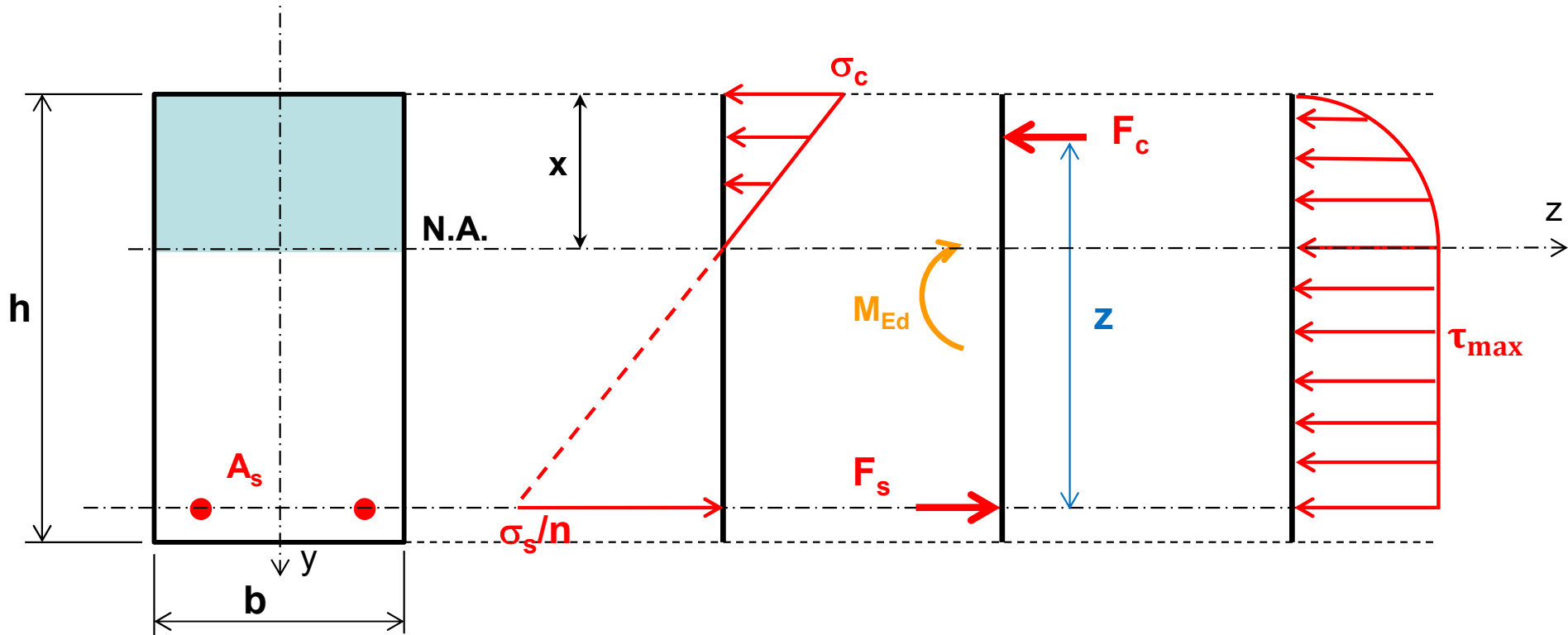


Rectangular section

$$\tau(y) = \frac{3}{2} \frac{V_y}{b h^3} (h^2 - 4y^2)$$

$$\tau_{\max} = \frac{3}{2} \frac{V_y}{S}$$

RC rectangular section at SLS (elasticity)



Static equilibrium

$$\mathbf{M}_{\text{Ed}} = \mathbf{F}_c \mathbf{z}$$

where $\mathbf{F}_c = \sigma_c \frac{\mathbf{b} \mathbf{x}}{2}$

$$\sigma_c = \frac{M_{Ed} \cdot x}{I_{hr}}$$

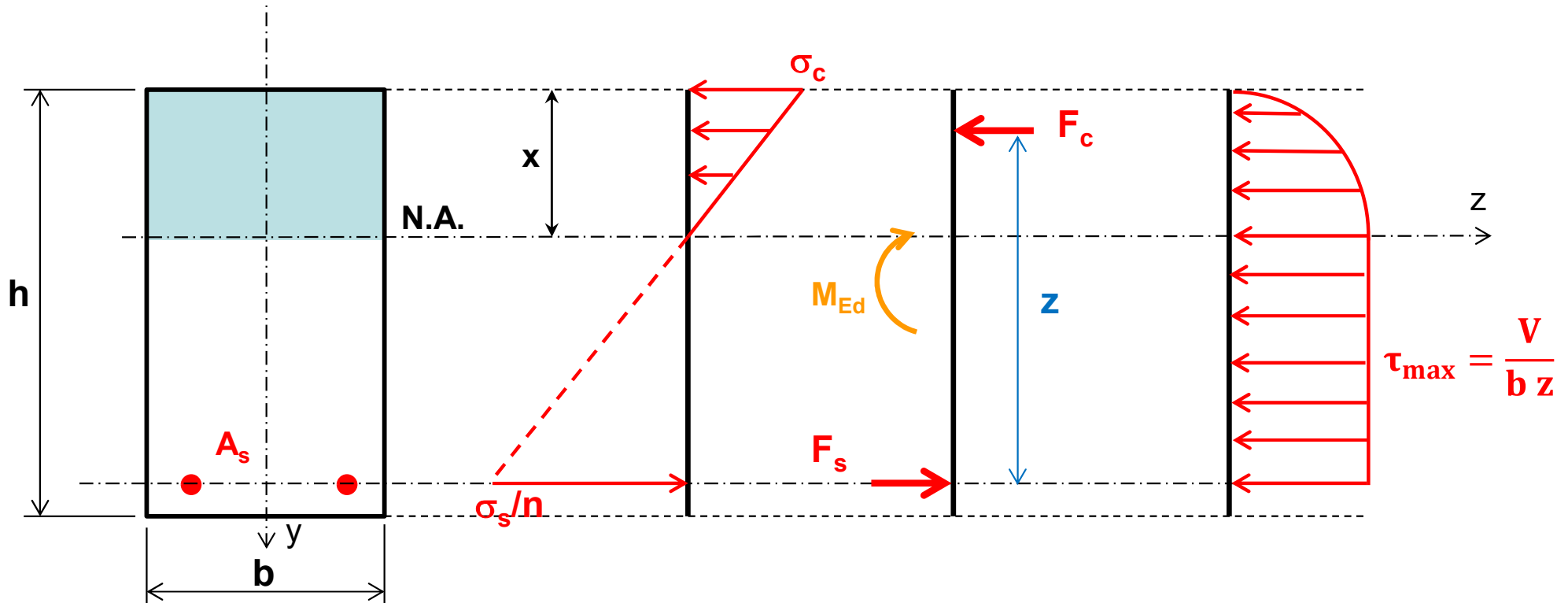
$$M_{Ed} = \frac{M_{Ed}}{I_{hr}} \frac{b x^2}{2} z$$

$= \mu_0 \rightarrow$ 1st order moment of area of concrete in compression

$$\mathbf{I}_{hr} = \mu_0 \mathbf{Z}$$

2 – Shear stress

RC rectangular section at SLS (elasticity)



$$I_{hr} = \mu_0 z$$

$$\tau(y) = \frac{V_y}{b(y)} \frac{\mu_z(y)}{I_{hr}}$$



$$\tau(y) = \frac{V}{z} \frac{\mu_z(y)}{\mu_0} \frac{1}{b}$$

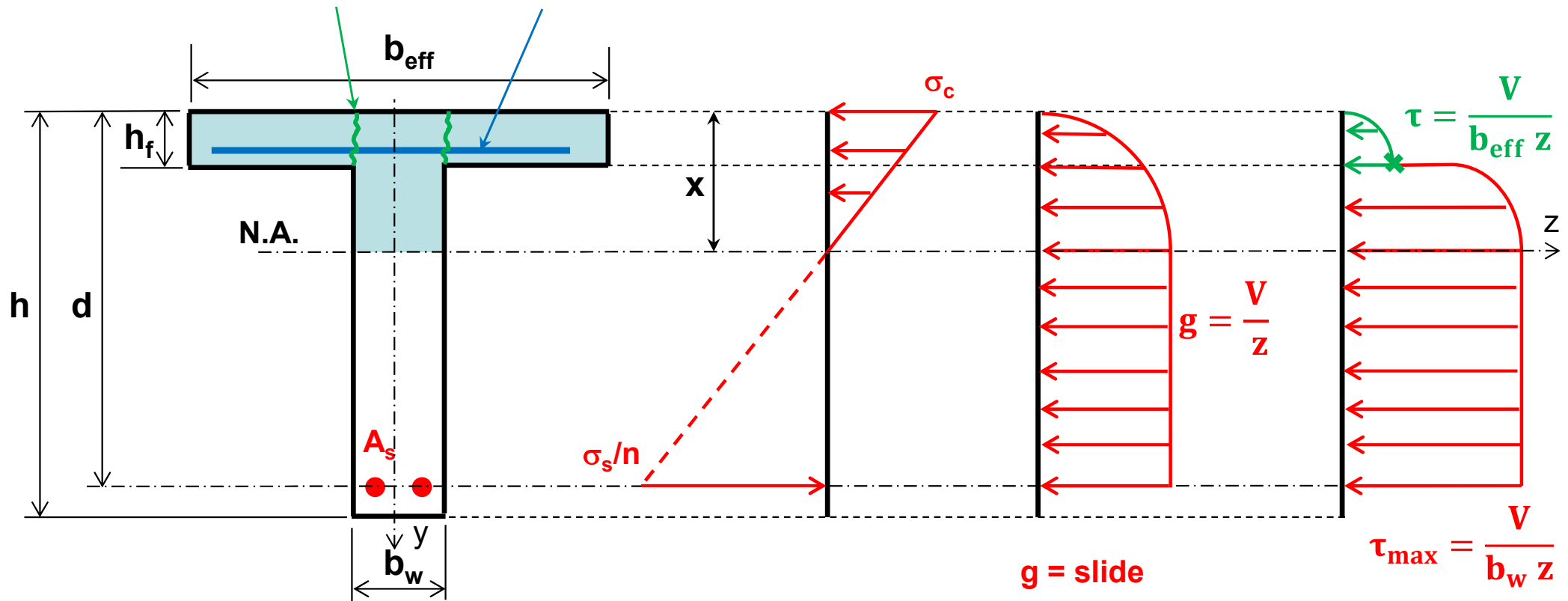
$$\tau_{max} = \frac{V}{b z}$$

2 – Shear stress

RC T-section at SLS (elasticity)

Risque de rupture entre
la table et la nervure

Armature de
coudre

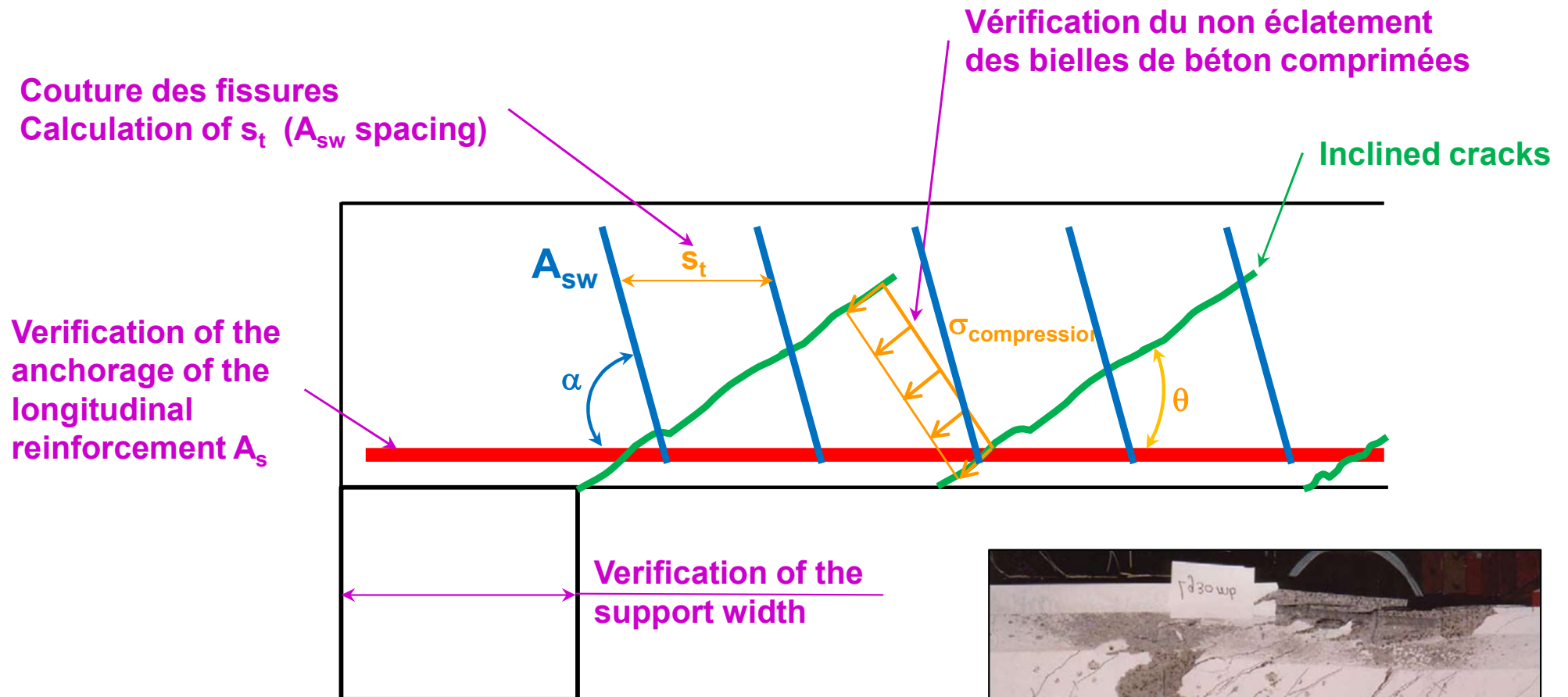


Effect of shear stress in a RC beam



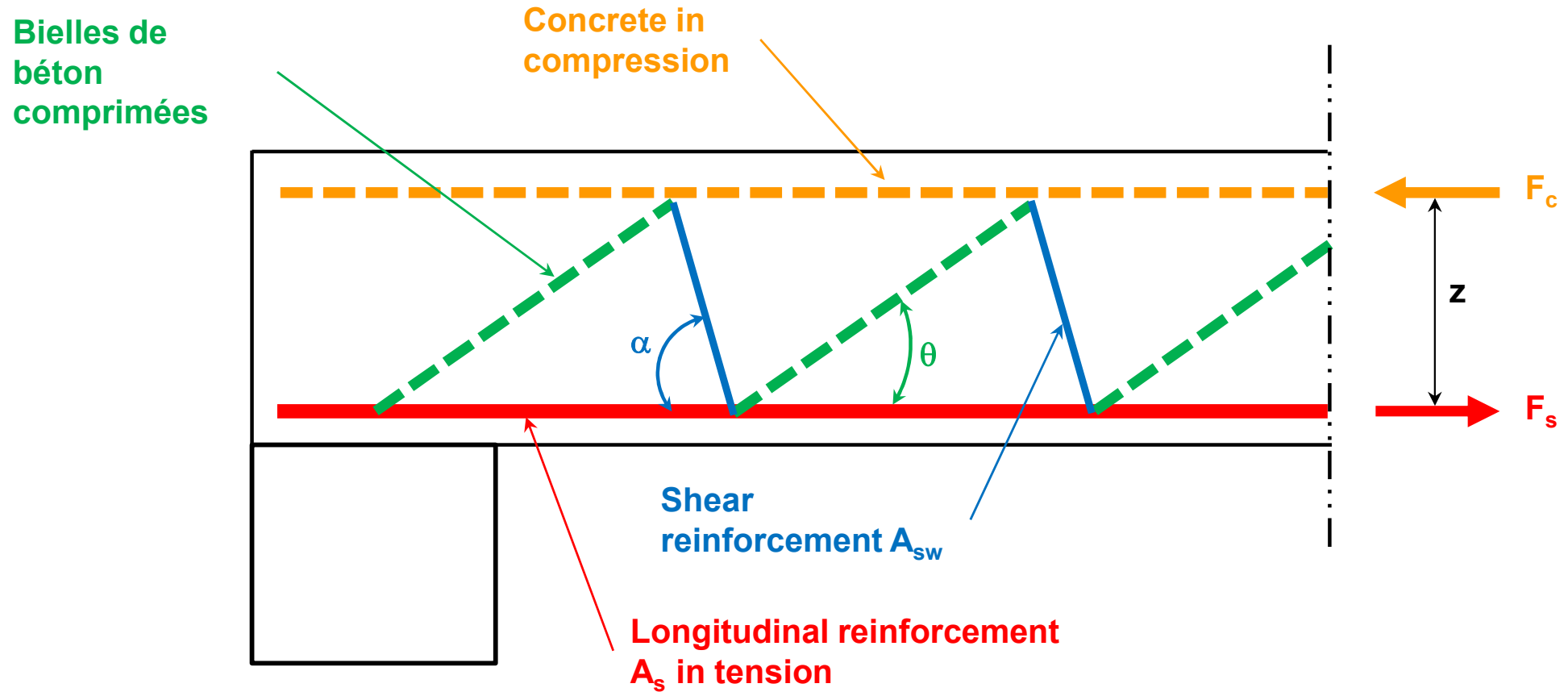
2 – Shear stress

Effect of shear stress in a RC beam



2 – Shear stress

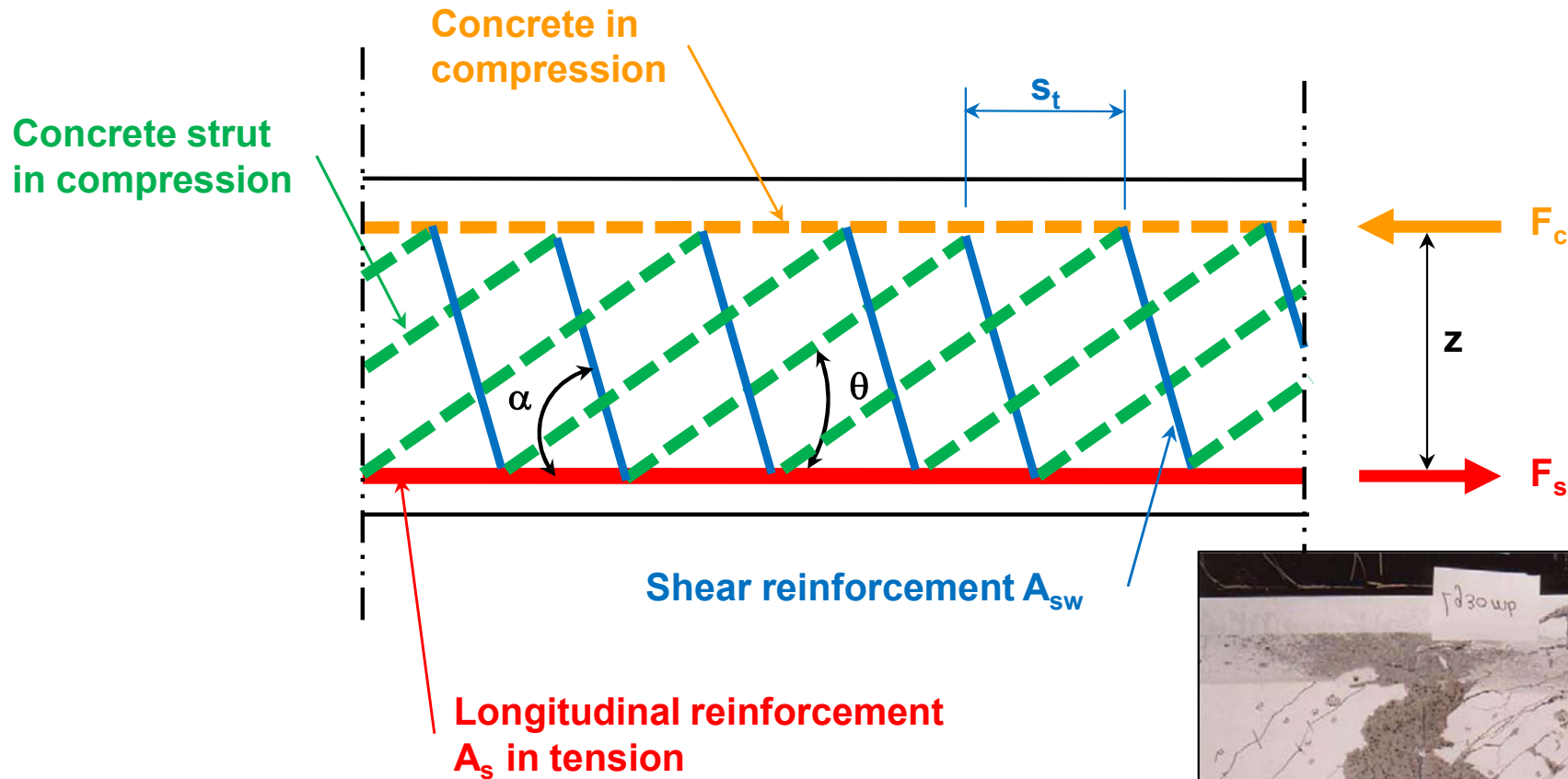
Ritter's and Mörsch's truss analogy model



2 – Shear stress

Truss model proposed in EC2

An inclined crack is stitched by several shear bars A_{sw} spaced by s_t



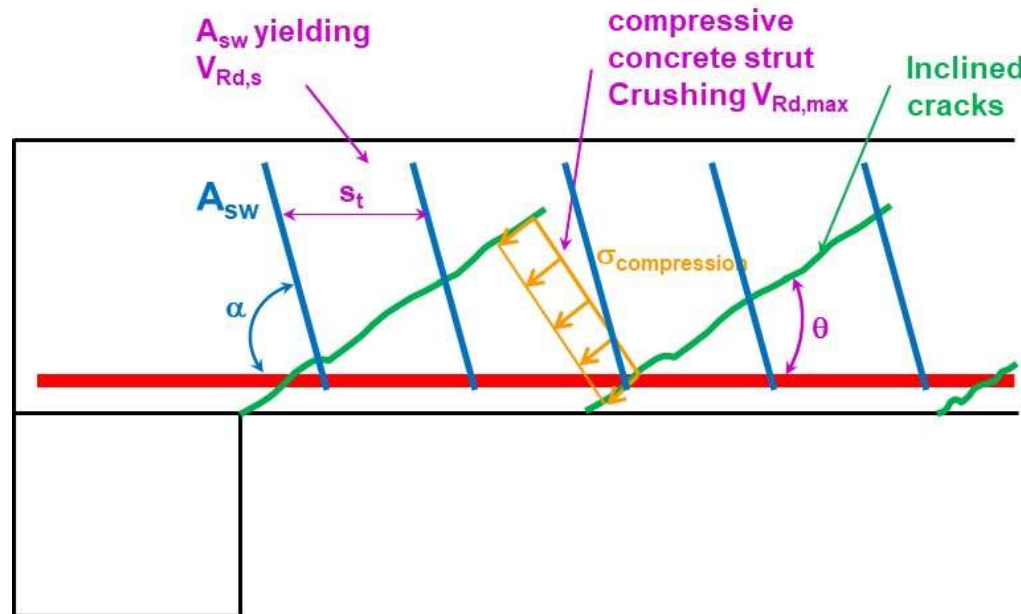
3 – Shear reinforcement verifications

Only design at ULS

➤ V_{Ed} : Design value of the shear force at ULS → $V_{Ed,r}$ or $V_{Ed,edge}$

3 different shear resistances:

- $V_{Rd,c}$: Design shear resistance of the member without shear reinforcement
- $V_{Rd,max}$: Design shear force limited by the crushing of the compression struts
- $V_{Rd,s}$: Design shear force sustained by the yielding shear reinforcement



3 – Shear reinforcement verifications

2 possibilities

$$V_{Ed,r} \leq V_{Rd,c}$$

Minimum cross-sectional area
of shear reinforcement

$$V_{Ed,r} \geq V_{Rd,c}$$

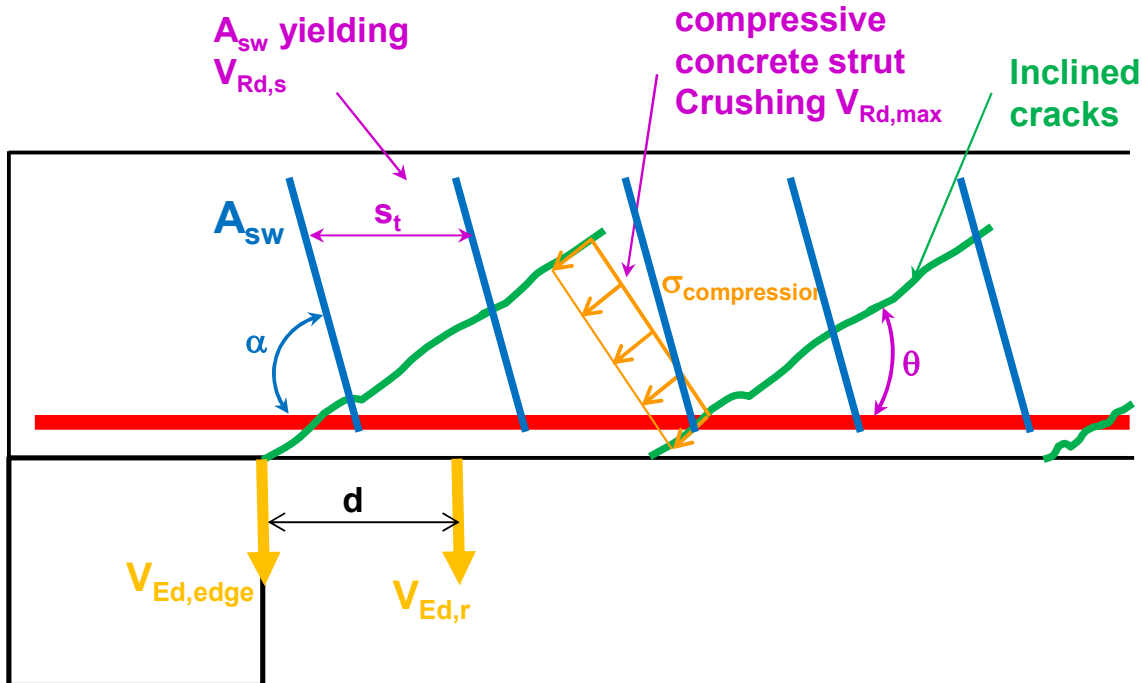
Design A_{sw}

Verifications

$$V_{Ed} \leq (V_{Rd,max} \text{ and } V_{Rd,s})$$

θ

s_t

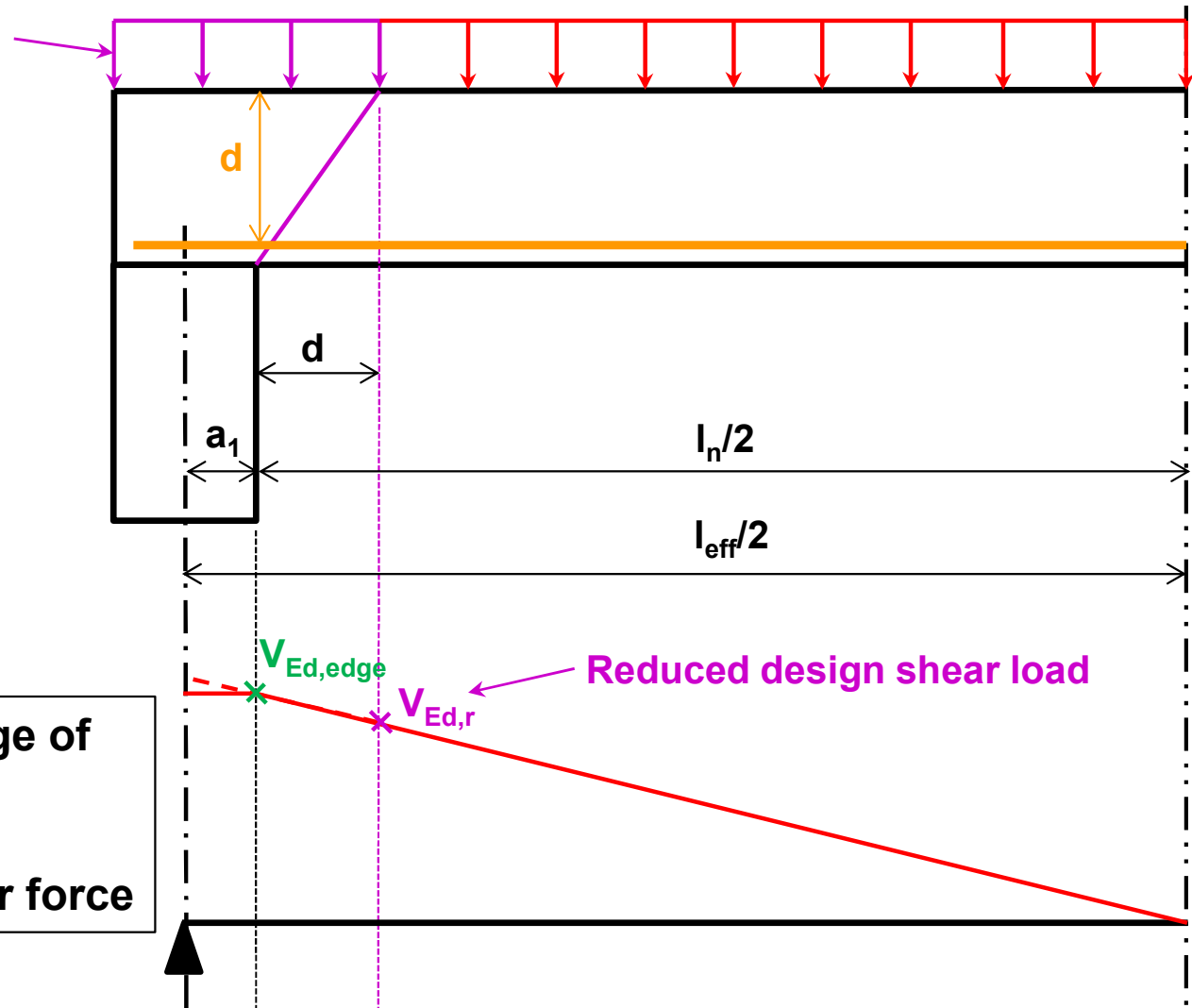


3 – Shear reinforcement verifications

Reduced design value $V_{Ed,r}$ of the shear force

Uniformly distributed loading

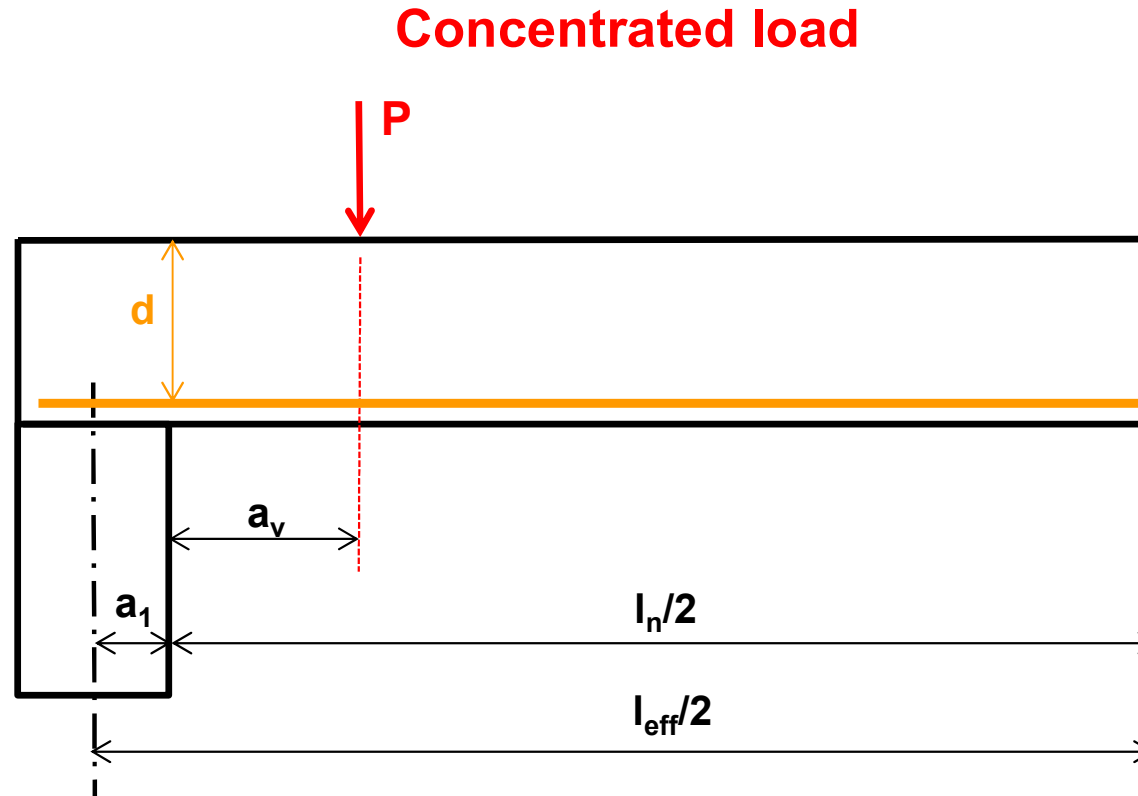
Charges transmises
directement à l'appui



$V_{Ed,edge}$ = shear force at the edge of the support

$V_{Ed,r}$ = reduced design shear force

3 – Shear reinforcement verifications



- | | |
|---------------------------|--------------------------------------|
| $a_v \geq 2 d$ | → No reduction of P |
| $0.5 d \leq a_v \leq 2 d$ | → P reduced by $\beta = a_v/(2 d)$ |
| $a_v \leq 0.5 d$ | → P reduced by $\beta = 0.25$ |

Can not be combined with the reduction due to uniformly concentrated loading

3 – Shear reinforcement verifications

Design shear resistance $V_{Rd,c}$

$V_{Rd,c}$ depends on f_{ck} , d , A_s , N_{Ed}

$$V_{Rd,c} = \left[\text{Max} \left\{ \underbrace{v_{\min}}; C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} \right\} + k_1 \sigma_{cp} \right] b_w d$$

Limiting shear stress of concrete alone

If $V_{Ed,r} \leq V_{Rd,c} \rightarrow$ Minimum cross-sectional area of shear reinforcement

Otherwise $\rightarrow$ Design shear reinforcement required

where $k = \text{Min} \left[1 + \sqrt{\frac{200}{d}}; 2 \right]$ (in mm)

$$k_1 = 0.15$$

$$C_{Rd,c} = \frac{0.18}{\gamma_c}$$

$$\sigma_{cp} = \text{Min} \left[\frac{N_{Ed}}{A_c}; 0.2 f_{cd} \right]$$

where A_c = area of concrete cross – section (in mm^2)

N_{Ed} = axial force due to loading or prestressing (in N)

$N_{Ed} = 0$ if tension and $\sigma_{cp} = 0$ if bending without axial load

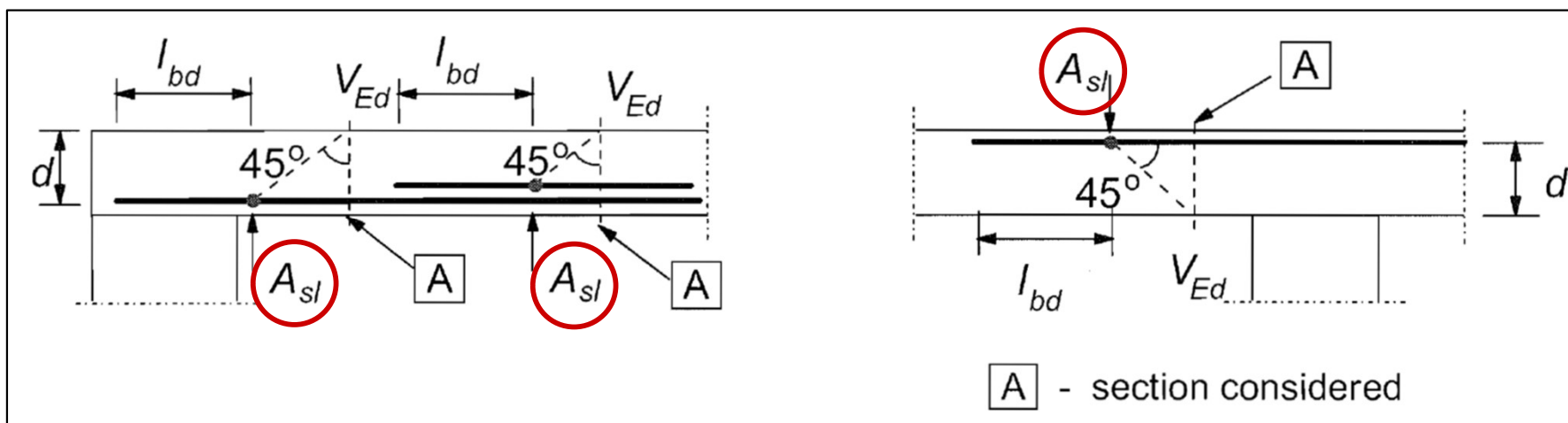
3 – Shear reinforcement verifications

$$v_{\min} = \frac{0.053}{\gamma_c} k^2 f_{ck}^{\frac{1}{2}} \quad \text{for beam}$$

$$v_{\min} = \frac{0.34}{\gamma_c} f_{ck}^{1/2} \quad \text{for slab}$$

$$v_{\min} = \frac{0.35}{\gamma_c} f_{ck}^{1/2} \quad \text{for wall}$$

$$\rho_1 = \frac{A_{sl}}{b_w d} \leq 0.02$$



NF EN 1992 (December 2004). Eurocode 2 – Design of concrete structures. CEN .

3 – Shear reinforcement verifications

Members requiring design shear reinforcement

$$V_{Ed,r} \geq V_{Rd,c}$$

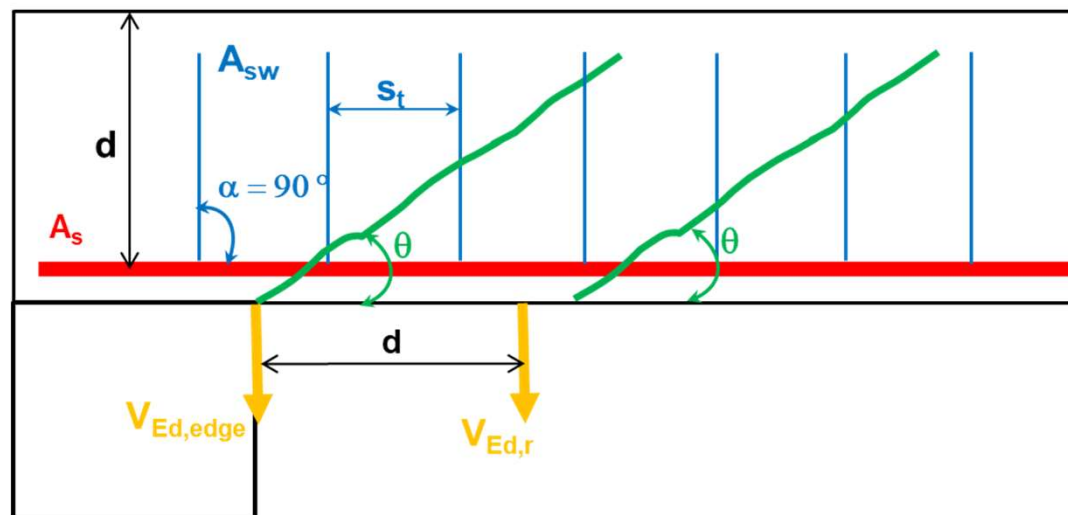
Methodology

Step 1: $V_{Ed,edge} \leq V_{Rd,max} \rightarrow$ Calculation of the **constant angle θ** of inclined struts for beam

Bending with or without compressive load $\rightarrow 21.8^\circ \leq \theta \leq 45^\circ \rightarrow 1 \leq \cot \theta \leq 2.5$

Step 2: $V_{Ed} \leq V_{Rd,s} \rightarrow$ Calculation of the **stirrups spacing s_t**

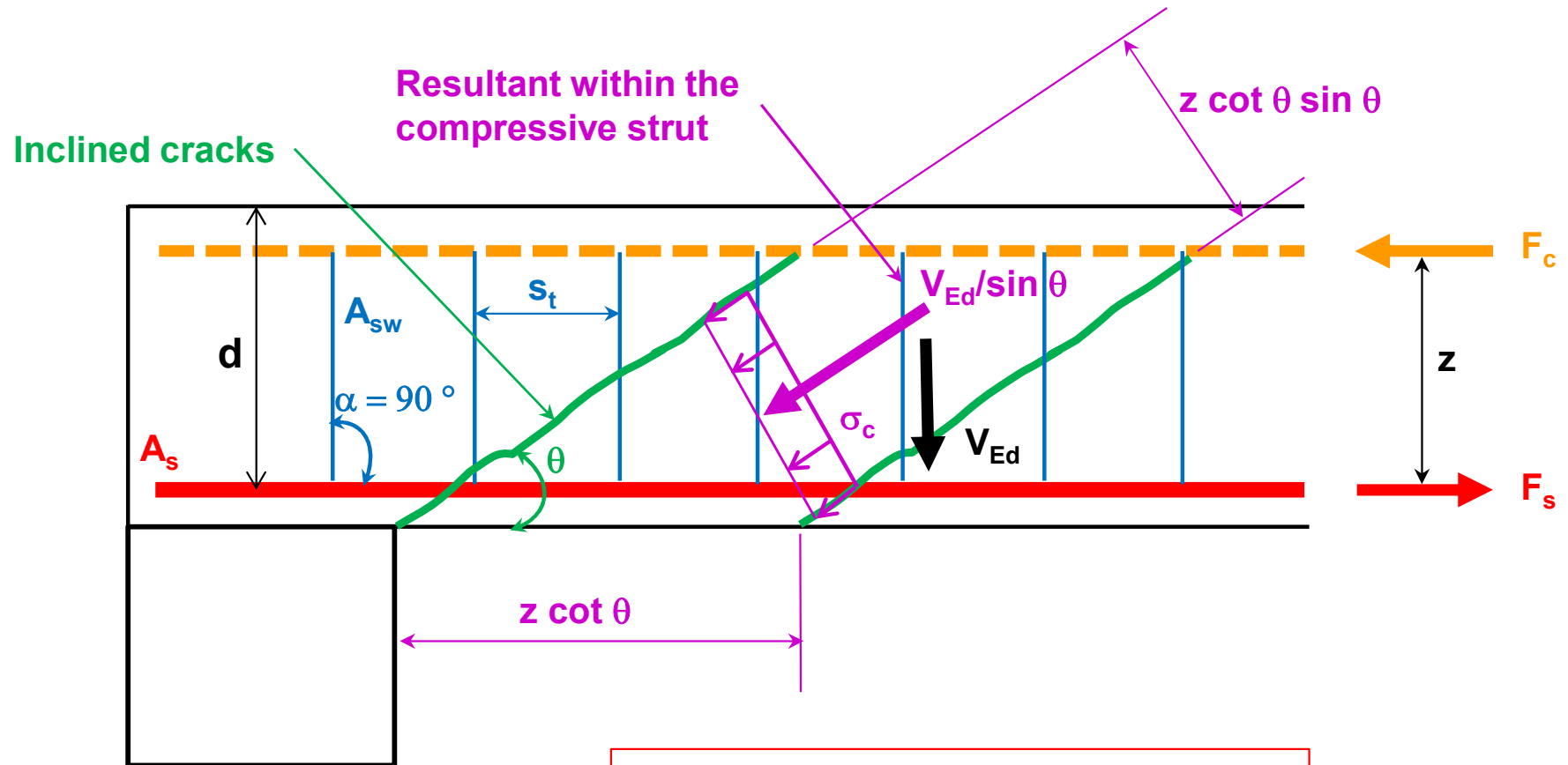
α = inclination angle of the stirrups $\rightarrow 45^\circ \leq \alpha \leq 90^\circ \rightarrow$ In practice $\alpha = 90^\circ$



3 – Shear reinforcement verifications

Inclination angle θ of the strut

Cross-section of a strut ($\alpha = 90^\circ$)

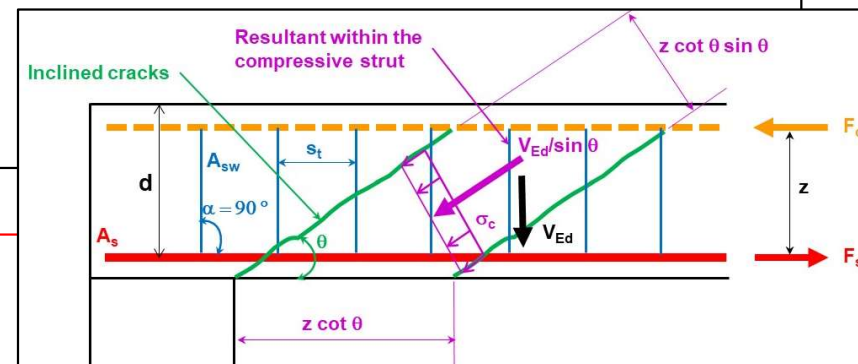


Bras de levier z	$= 0.9 d$
Strut thickness	$= \text{beam width } b_w$
Strut width	$= z \cot \theta \sin \theta$

3 – Shear reinforcement verifications

Inclination angle θ of the strut

$$\frac{V_{Rd,max}}{\sin \theta} = \underbrace{\alpha_{cw} v_1 f_{cd}}_{\text{Limiting compressive stress}} \underbrace{b_w z \cot \theta}_{\text{Strut cross-sectional area}} \sin \theta \quad \text{and} \quad V_{Ed} = V_{Ed,edge} \leq V_{Rd,max}$$



Minimum value θ_{min} of θ

$$V_{Ed,edge} \leq \alpha_{cw} v_1 f_{cd} b_w z \frac{\cot \theta}{1 + (\cot \theta)^2} \rightarrow \theta_{min} \rightarrow \theta_{min} \leq \theta \leq 45^\circ \text{ and } \theta \geq 21.8^\circ$$

where v_1 = Strength reduction factor for concrete cracked in shear

$$v_1 = v = 0.6 \left(1 - \frac{f_{ck}}{250} \right)$$

Several possible values for $\theta \rightarrow$ If θ decrease:

- The amount of shear reinforcement decrease
- The amount of longitudinal reinforcement increase
- The compressive stress within the strut increase

3 – Shear reinforcement verifications

Inclination angle θ of the strut

Value of α_{cw}

Loading		α_{cw}
Bending without axial load		1
Bending with compression load Prestressed structures	$0 < \sigma_{cp} \leq 0,25 \cdot f_{cd}$	$1 + \frac{\sigma_{cp}}{f_{cd}}$
	$0,25 \cdot f_{cd} < \sigma_{cp} \leq 0,5 \cdot f_{cd}$	1,25
	$0,5 \cdot f_{cd} < \sigma_{cp} < 1,0 \cdot f_{cd}$	$2,5 \cdot \left(1 - \frac{\sigma_{cp}}{f_{cd}}\right)$
Bending with tension load (French national annex)		$1 + \frac{\sigma_{ct}}{f_{ctm}}$
σ_{cp} is the mean compressive stress, measured positive, in the concrete due to the design axial force. This should be obtained by averaging it over the concrete section taking account of the reinforcement. The value of σ_{cp} need not be calculated at a distance less than $0.5d \cot \theta$ from the edge of the support.		

D. Ricotier (2012). Dimensionnement des structures en béton selon l'Eurocode 2. Edition Le Moniteur, Paris, France.

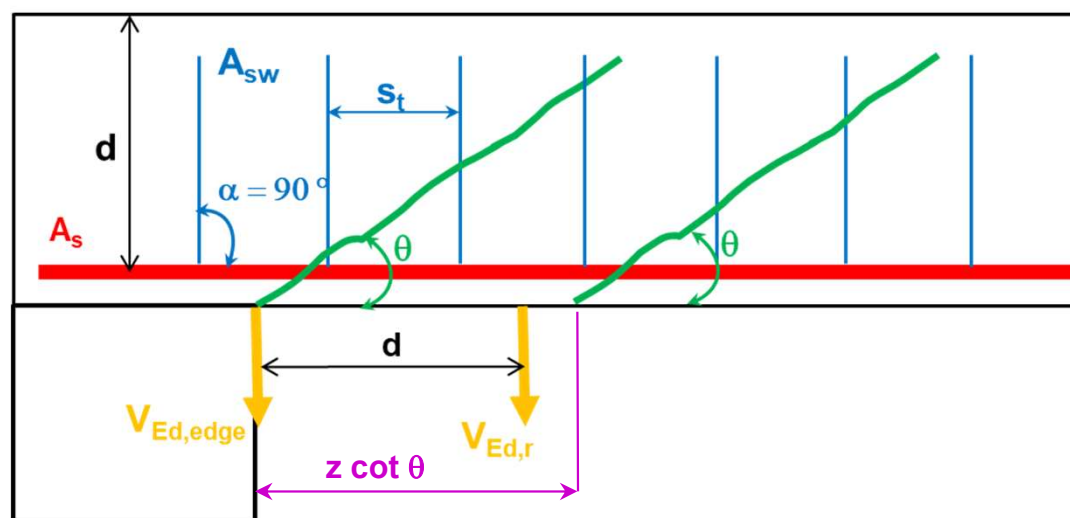
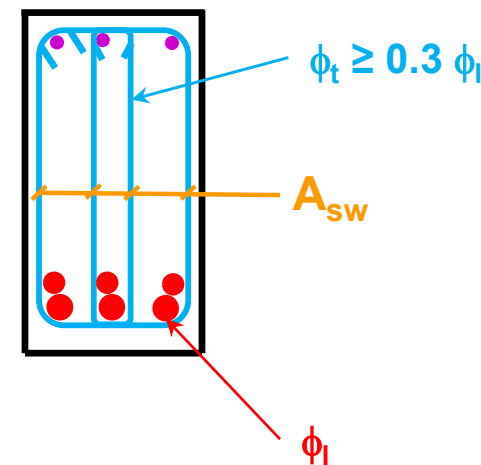
3 – Shear reinforcement verifications

Spacing s_t of shear reinforcement at supports

$$V_{Ed,r} \leq V_{Rd,s} = \underbrace{\frac{z \cot \theta}{s_t}}_{\text{Number of } A_{sw} \text{ string crossing a crack}} A_{sw} f_{yd} \quad \text{where } z = 0.9 d$$

Number of A_{sw} string crossing a crack

$$s_t \leq \frac{A_{sw}}{V_{Ed,r}} z \cot \theta f_{yd}$$



3 – Shear reinforcement verifications

Minimum ratio and maximum longitudinal spacing
of shear reinforcement required along the beam

Minimum ratio ρ_w for beam

$$\rho_w = \frac{A_{sw}}{b_w s_t} \geq \rho_{min} = \frac{0.08 \sqrt{f_{ck}}}{f_{yk}} \rightarrow s_t \leq \frac{f_{yk}}{0.08 \sqrt{f_{ck}}} \frac{A_{sw}}{b_w}$$

Maximum longitudinal spacing $s_{t,max}$

$$s_t \leq s_{t,max} = \text{Min}[0.75 d ; 600 \text{ mm}]$$

If compressive reinforcement (buckling) $\rightarrow s_t \leq 15 \phi_{A's}$

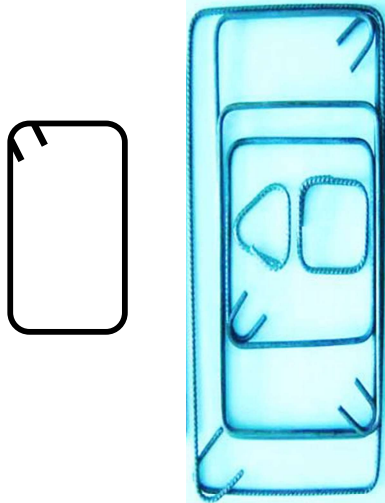
$\phi_{A's} = A'_s$ minimum diameter

3 – Shear reinforcement verifications

Shear reinforcement detailing

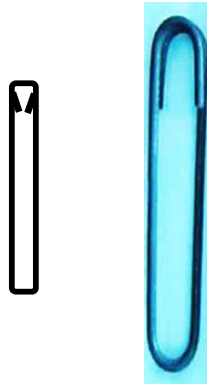
Closed stirrup

Cadre



2 legged stirrup

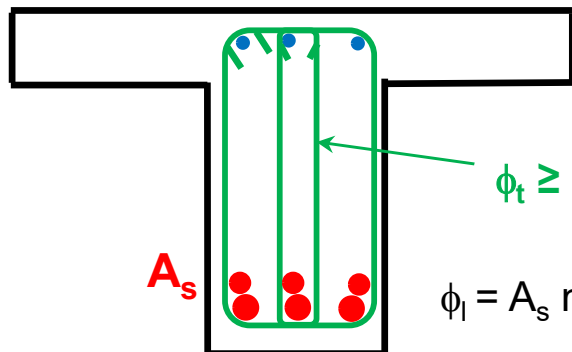
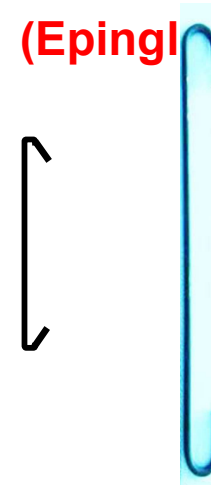
(Etrier)



Opened stirrup

Single legged stirrup

(Epingl)



$$\phi_t \geq 0.3 \phi_l$$

$\phi_l = A_s$ maximum diameter

Rebar diameter ϕ_t

Between 6 and 12 mm

3 – Shear reinforcement verifications

Detailing arrangement of shear reinforcement along the beam

Uniformed distributed loading

From each side of the beam

Step 1: 1st strut

Calculate s_{t0} with $V_{Ed,r}$

Step 2 : Following struts

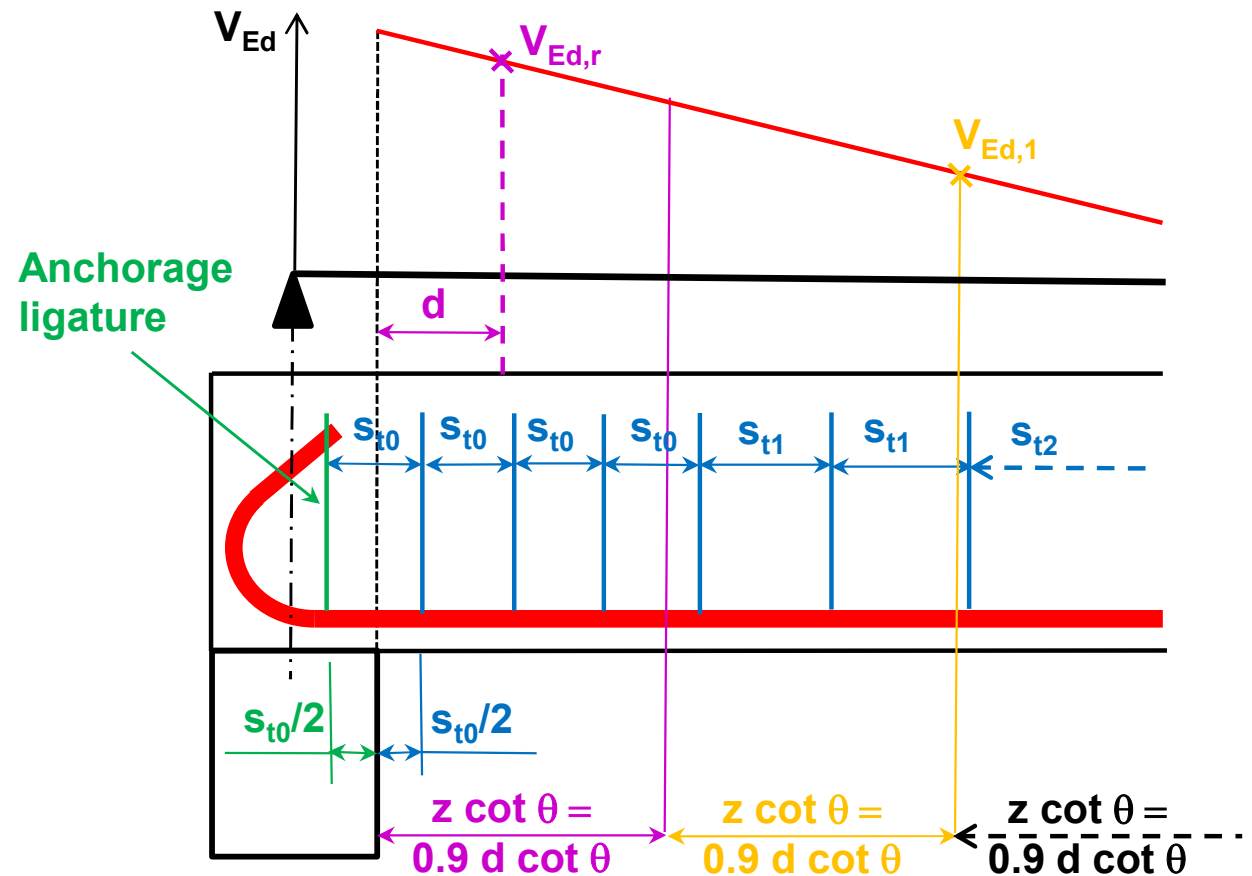
Calculable s_t for each

length increment ($z \cot \theta$)

with the smallest value of V_{Ed}

in the increment

Repeat until $V_{Ed} = 0$



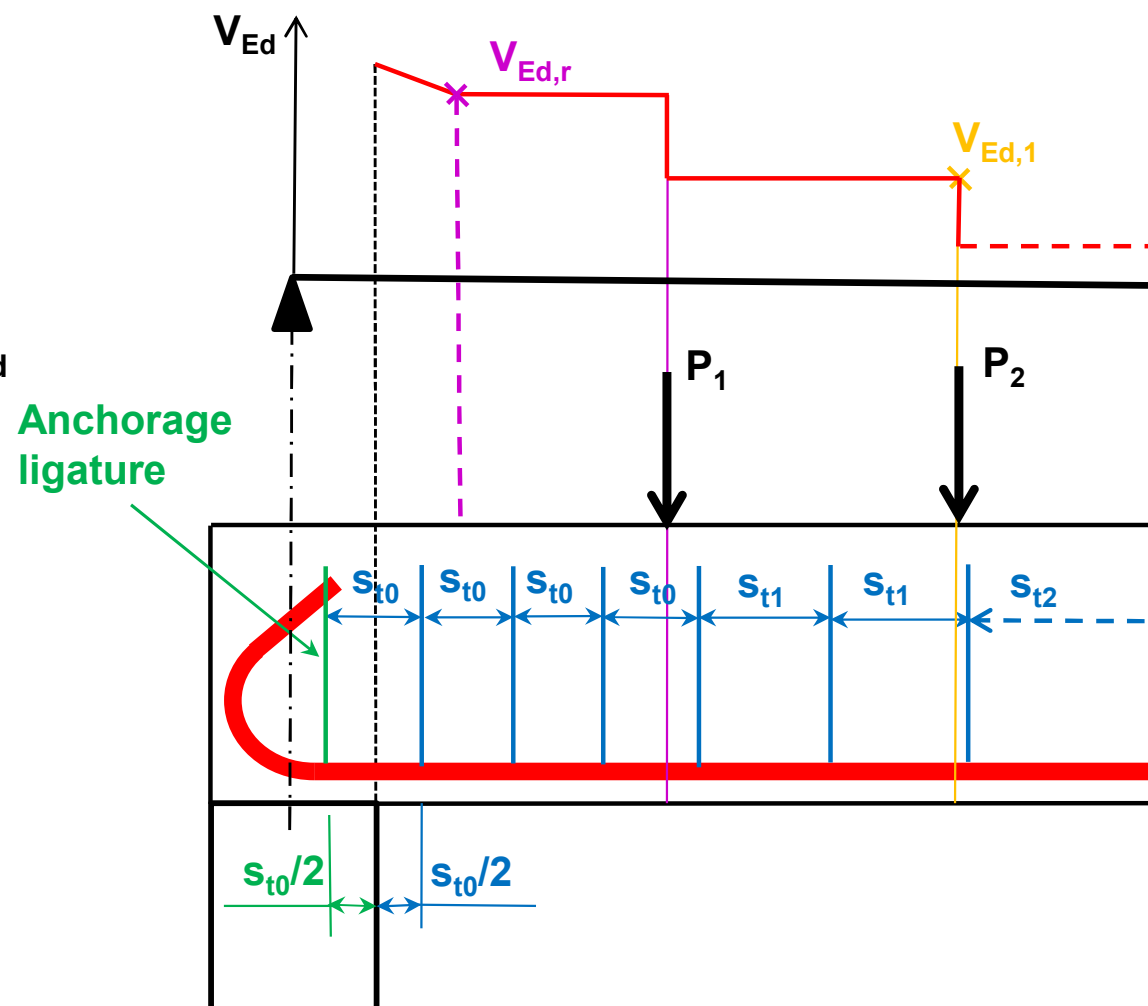
3 – Shear reinforcement verifications

Detailing arrangement of shear reinforcement along the beam

Between the point loads

- Constant design shear force V_{Ed}
- Constant spacing s_t
- First spacing s_{t0} calculated with $V_{Ed,r}$

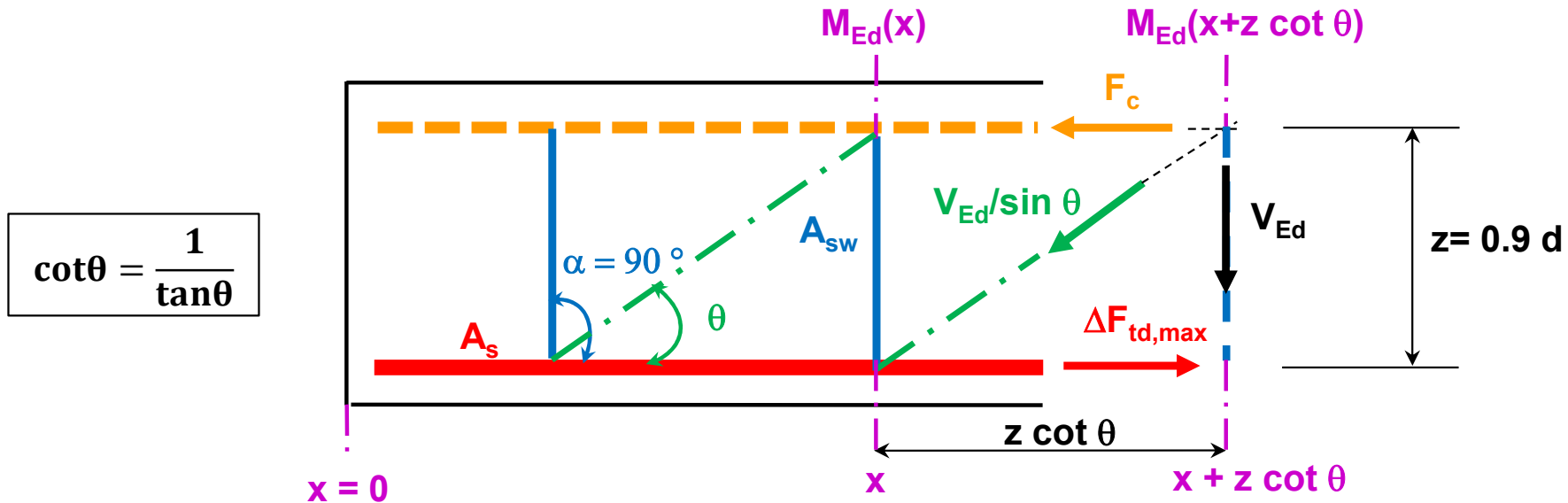
Concentrated loads



4 – Curtailment of longitudinal reinforcement

Effect of shearing on longitudinal reinforcement

Ritter's and Mörsch's truss model (single truss)



Additional tensile force in A_s : $\Delta F_{td,max} = V_{Ed} \cot \theta = \frac{M_{Ed}(x + z \cot \theta)}{z}$

Shift of the moment curve : $z \cot \theta$

EC2 truss model (multiple trusses)

Additional tensile force in A_s

ranging from 0 to $\Delta F_{td,max}$

→ A mean value used: $\Delta F_{td} = \frac{\Delta F_{td,max}}{2} = \frac{V_{Ed} \cot \theta}{2}$

EC2 shift of the moment curve

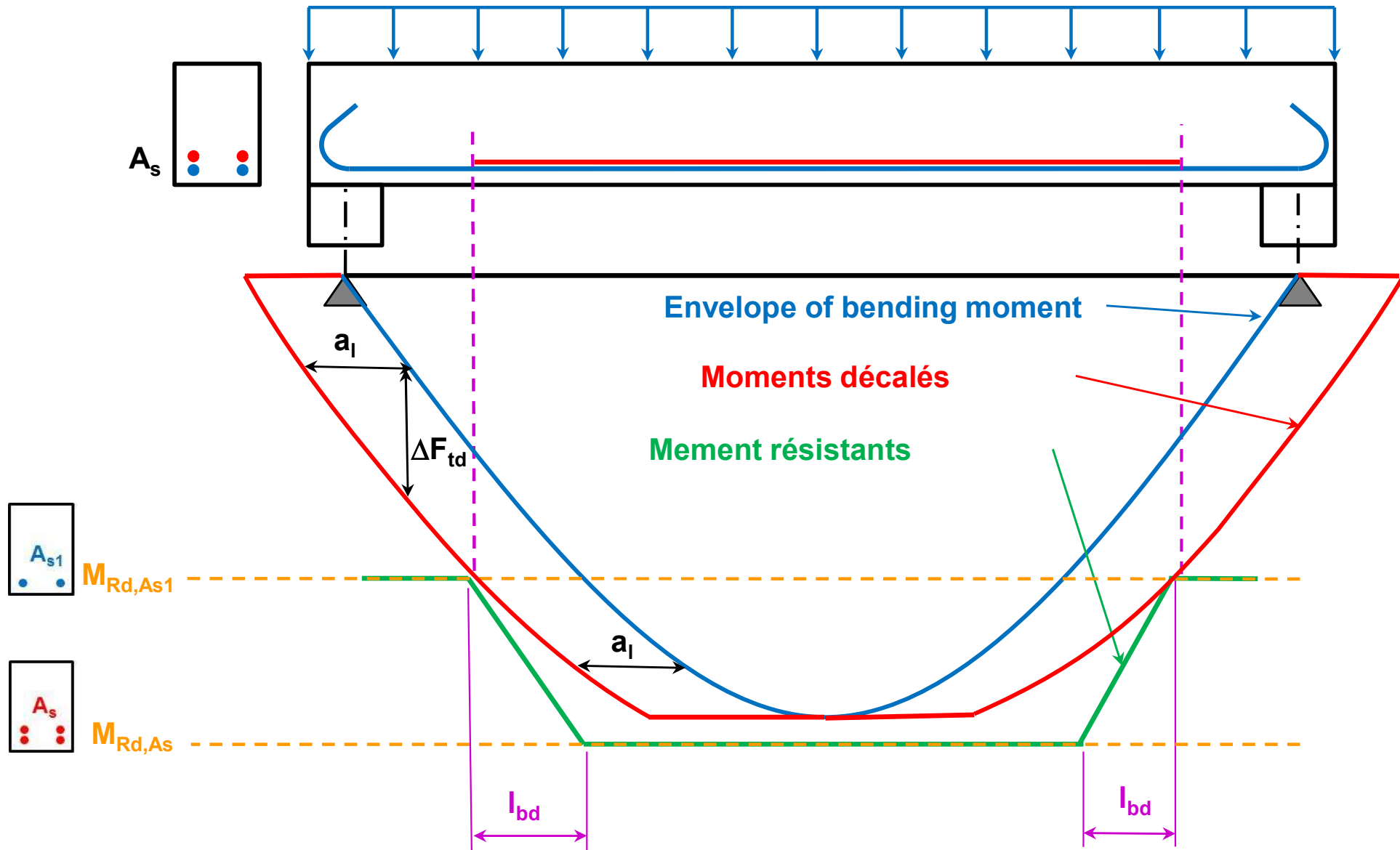
$$a_l = \frac{z}{2} \cot \theta$$

Member without shear reinforcement

$$a_l = d$$

4 – Curtailment of longitudinal reinforcement

Effect of shearing on longitudinal reinforcement



l_{bd} = design anchorage length

4 – Curtailment of longitudinal reinforcement

Effect of shearing on longitudinal reinforcement

Methodology

1. Plot the envelope bending moment curve M_{Ed} at ULS
2. Décaler la courbe de $a_l = \frac{z}{2} \cot \theta$ dans le sens défavorable

For member without shear reinforcement $a_l = d$

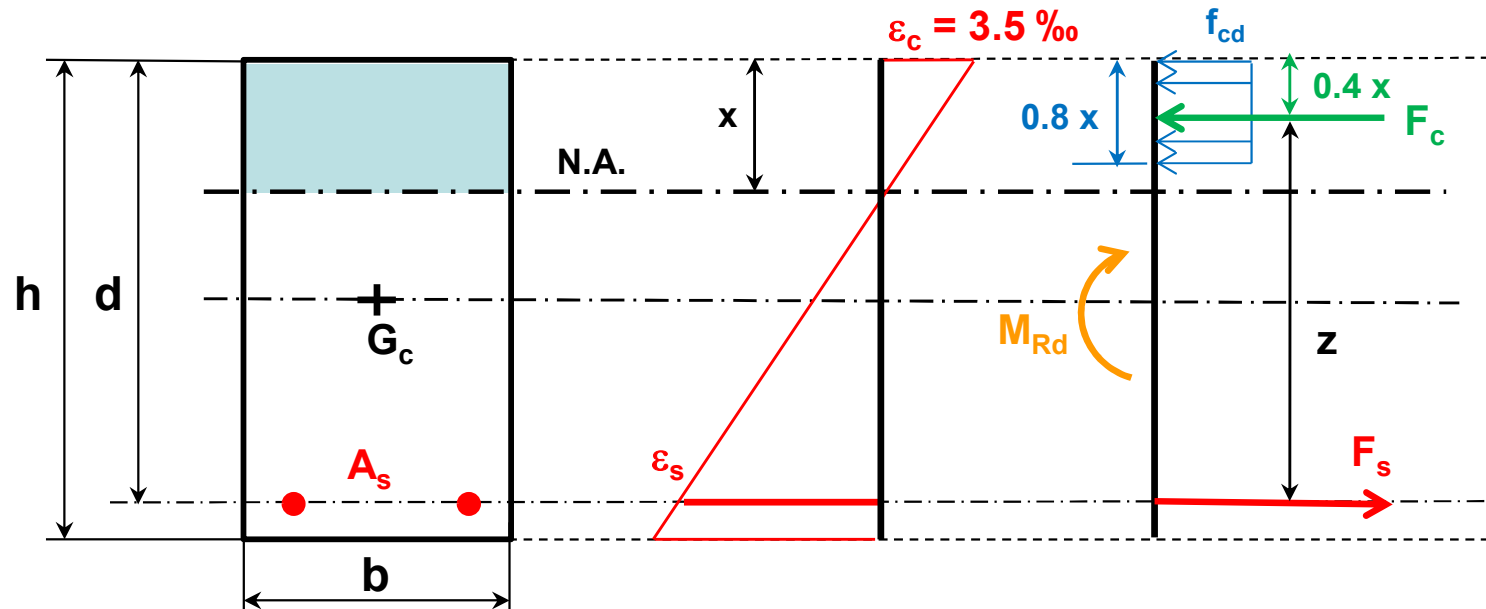
3. Calculate the resisting bending moments M_{Rd} for RC sections with 1 layer, 2 layers and 3 layers of longitudinal reinforcement A_s
4. Trouvez les points où $M_{Rd} = M_{Ed} \rightarrow$ Curtail the 2nd and 3rd layers of longitudinal reinforcement (épure d'arrêt des barres)
5. Verify the required design anchorage length l_{bd} between 2 layers

4 – Curtailment of longitudinal reinforcement

Resisting bending moment M_{Rd}

Assumptions: $\varepsilon_c = 3.5 \text{ ‰}$ and $\sigma_s = 435 \text{ MPa}$

Rectangular section



Static equilibrium

$$\begin{cases} F_s - F_c = 0 \\ M_{Rd} = F_c z \end{cases}$$

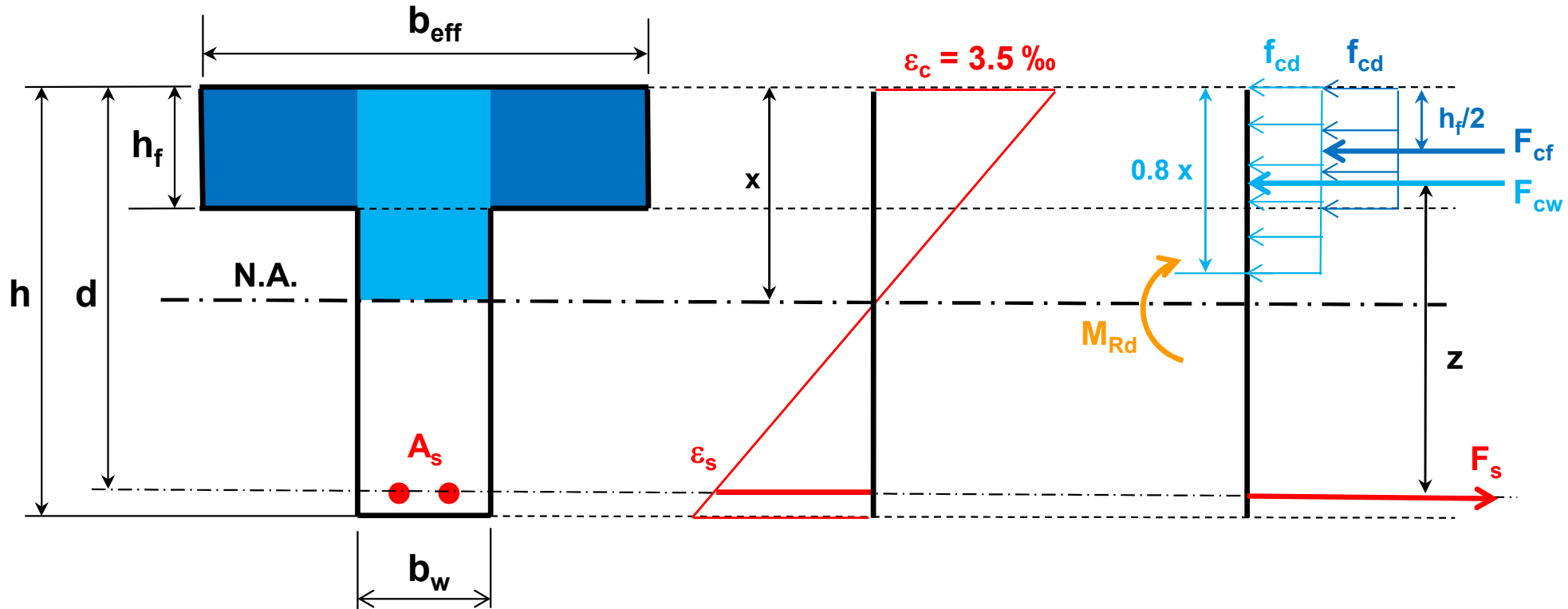


$$M_{Rd} = A_s f_{yd} \left(d - \frac{A_s f_{yd}}{2 b f_{cd}} \right)$$

4 – Curtailment of longitudinal reinforcement

Resisting bending moment M_{Rd}

T-Section



Static equilibrium

$$\begin{cases} F_s - F_{cf} - F_{cw} = 0 \\ M_{Rd} = F_{cw} z + F_{cf} \left(d - \frac{h_f}{2} \right) \end{cases}$$

$$M_{Rd} = 0.8 x b_w f_{cd} (d - 0.4 x) + (b_{eff} - b_w) h_f f_{cd} \left(d - \frac{h_f}{2} \right)$$

$$x = \frac{A_s f_{yd} - (b_{eff} - b_w) h_f f_{cd}}{0.8 b_w f_{cd}} \geq h_f \text{ if not, rectangular section } (b_{eff} h)$$

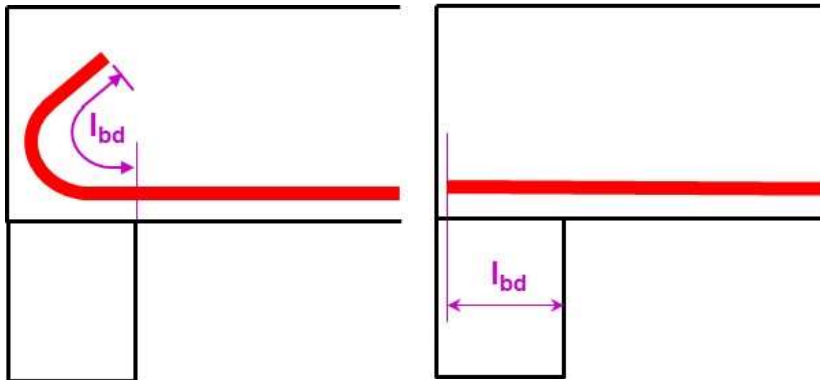
5 – Verifications at supports

EC2 truss model

End supports

$$\Delta F_{td} = V_{Ed,edge} \frac{\cot \theta}{2}$$

$$A_s \geq \frac{V_{Ed,edge}}{f_{yd}} \frac{\cot \theta}{2}$$



l_{bd} = design anchorage length

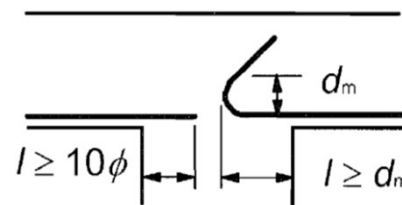
Intermediate supports

$$\Delta F_{td} = V_{Ed,edge} \frac{\cot \theta}{2} - \frac{|M_{Ed}|}{z}$$

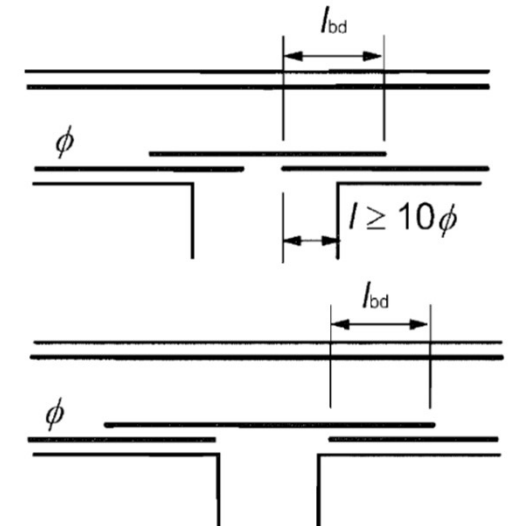
$$A_s \geq \frac{1}{f_{yd}} \left(V_{Ed,edge} \frac{\cot \theta}{2} - \frac{|M_{Ed}|}{z} \right)$$

where M_{Ed} = Negative moment at intermediate support

z = Inner lever arm



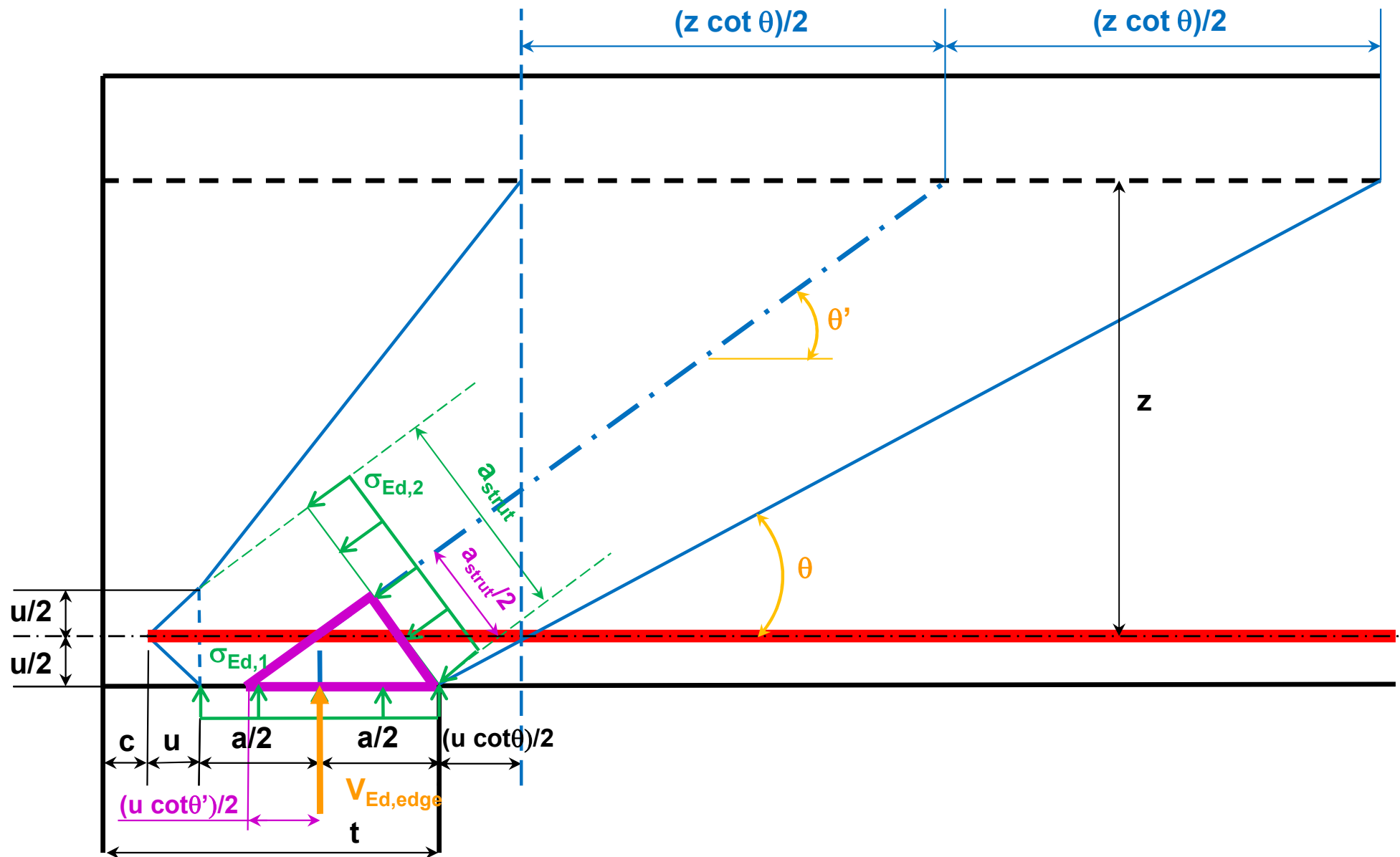
d_m = diameter of the mandrel
 ϕ = diameter of the rebar



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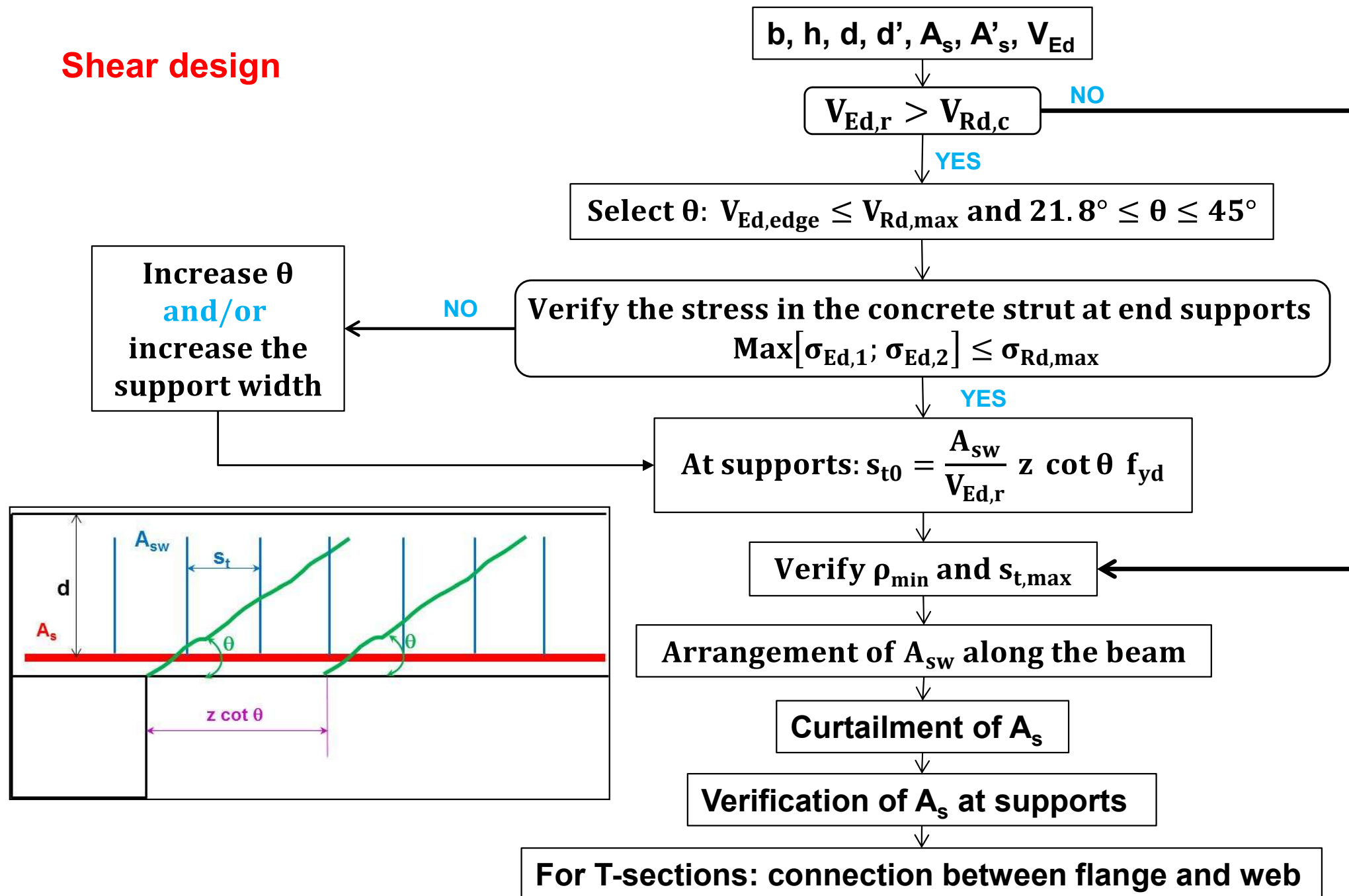
5 – Verifications at supports

$$b_{\text{supp}} = \text{beam width at support and } a_{\text{strut}} = 2 \frac{u \cot \theta' + a}{2} \sin \theta' = u \cos \theta' + a \sin \theta'$$



5 – Verifications at supports

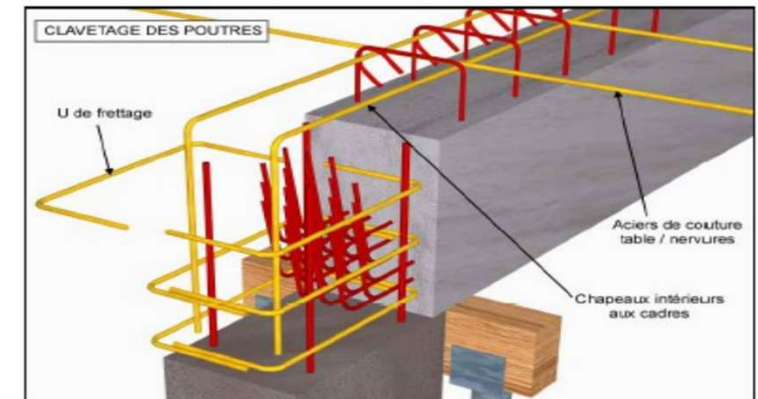
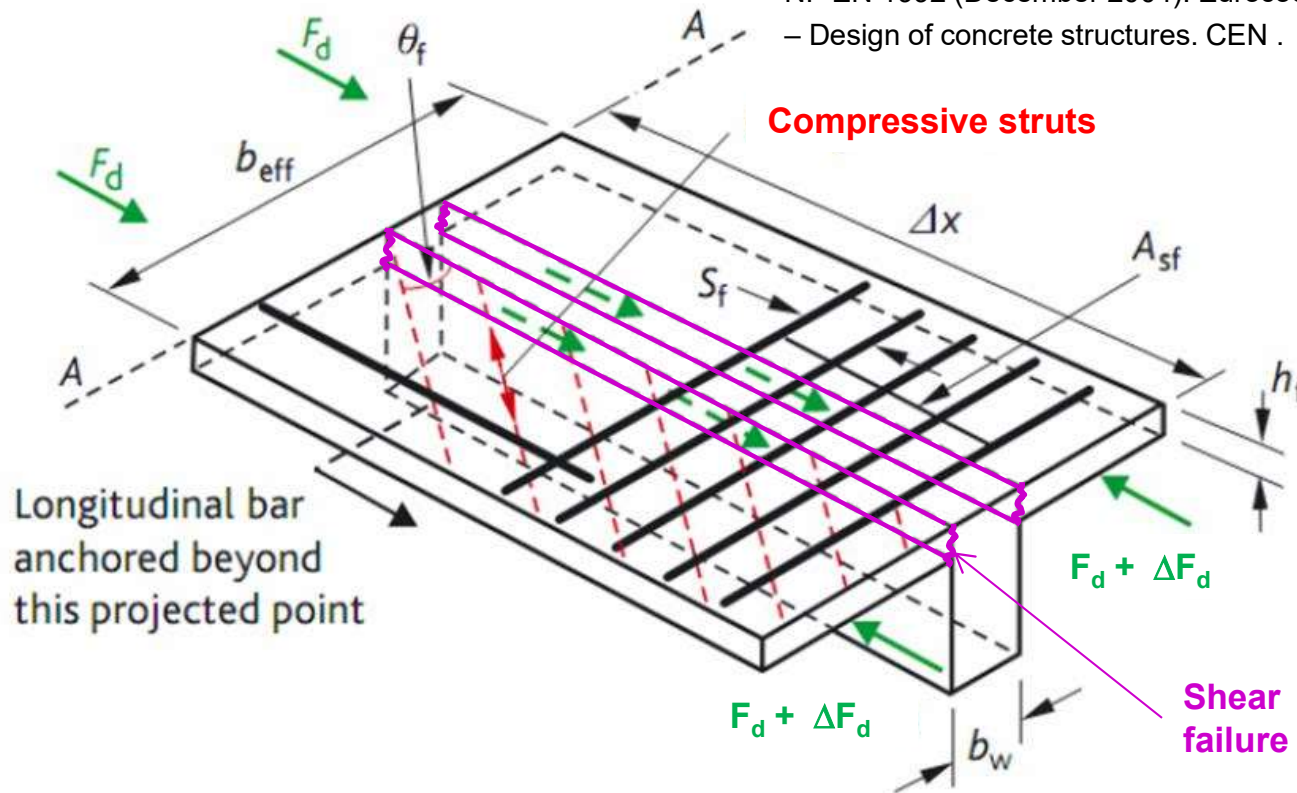
Shear design



6 – Flange and web connection

Minimum area of transverse reinforcement A_{sf}

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ΔF_d = Variation of the normal force in the flange over the length Δx

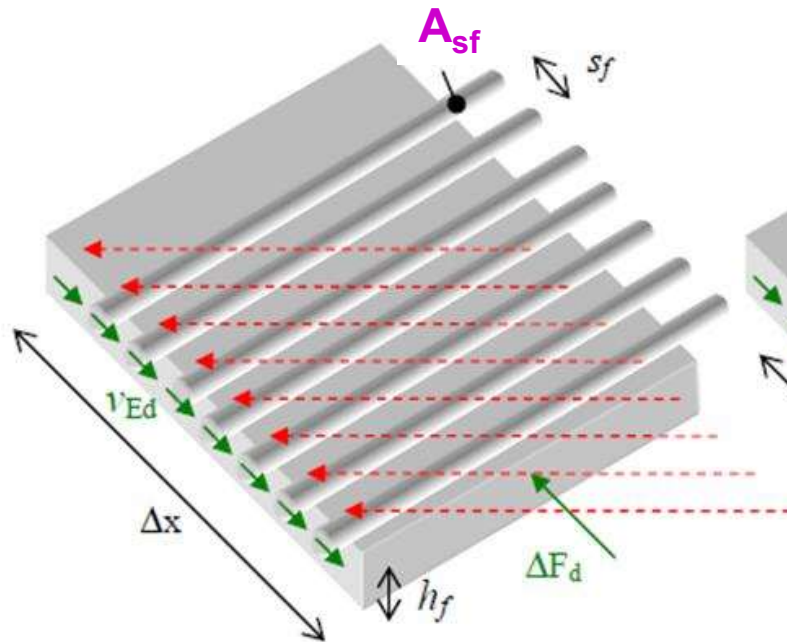
A_{sf} = Cross sectional area of transverse reinforcement (m^2)

s_f = Spacing of transverse reinforcement (m)

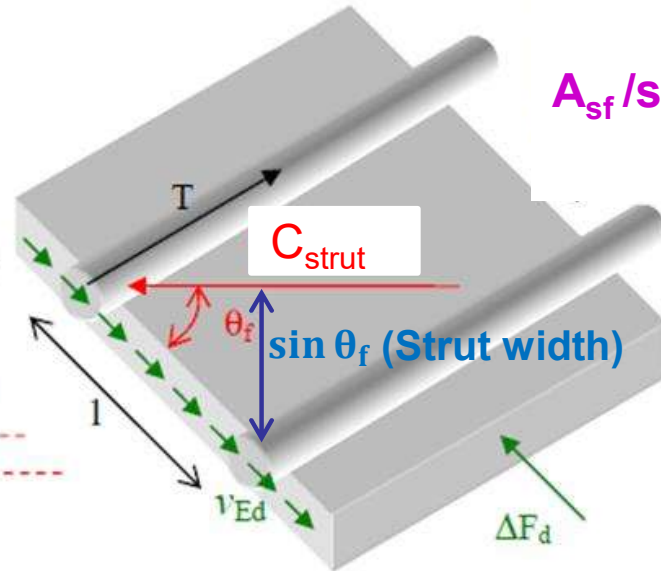
θ_f = Inclination angle between the transverse struts within the flange and the beam axis
($26.5^\circ \leq \theta_f \leq 45^\circ$)

6 – Flange and web connection

Compressive stress within the strut

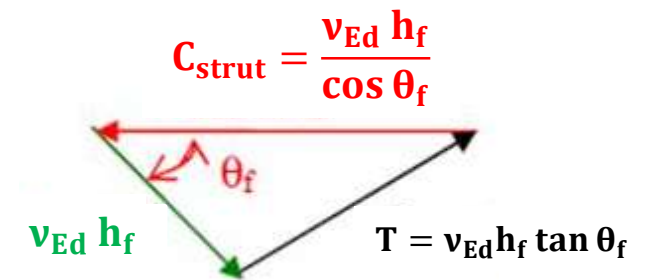


Flange web junction



Unit mesh model

C_{strut} = compressive force within strut
 A_{sf} = transverse reinforcement area
 T = tension force in A_{sf}
 v_{Ed} = longitudinal shear stress



Equilibrium of forces

P. Boeraeve (2011). Manuel de béton armé. Institut Gramme – Unité de construction

Longitudinal shear stress

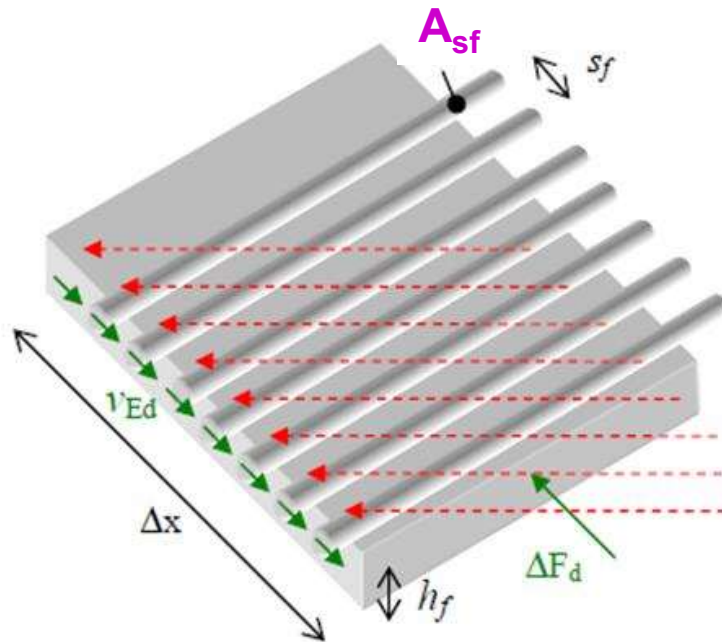
$$: v_{Ed} = \frac{\Delta F_d}{h_f \Delta x}$$

Compressive stress within the strut : $\sigma_{strut} = \frac{v_{Ed}}{\cos \theta_f} \frac{1}{\sin \theta_f}$

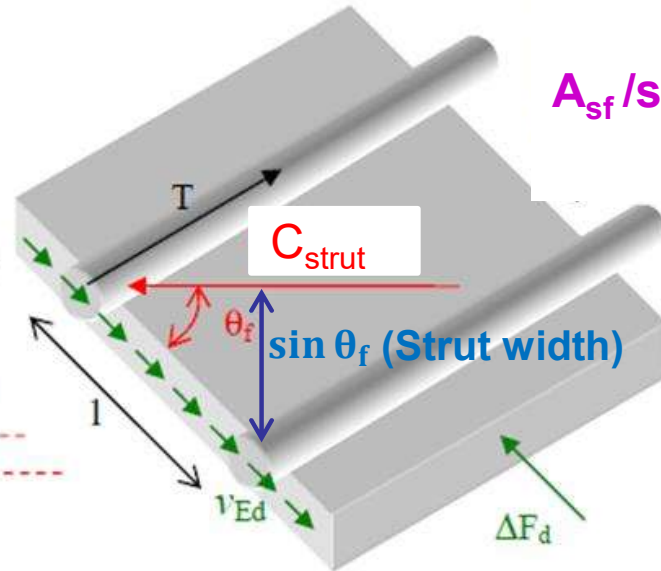
Design strength to prevent crushing : $v_{Max} = v f_{cd}$ where $v = 0.6 \left(1 - \frac{f_{ck}(\text{MPa})}{250} \right)$

6 – Flange and web connection

Inclination angle of the strut θ_f

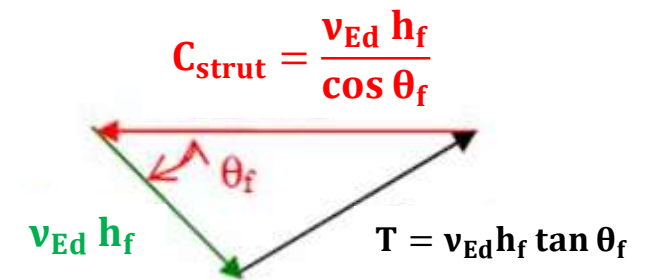


Flange web junction



Unit mesh model

C_{strut} = compressive force within strut
 A_{sf} = transverse reinforcement area
 T = tension force in A_{sf}
 v_{Ed} = longitudinal shear stress



Equilibrium of forces

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$$\sigma_{strut} \leq v_{Max} \rightarrow \frac{v_{Ed}}{\cos \theta_f} \frac{1}{\sin \theta_f} \leq v f_{cd} \rightarrow v_{Ed} \leq v f_{cd} \sin \theta_f \cos \theta_f = \frac{1}{2} v f_{cd} \sin(2\theta_f)$$

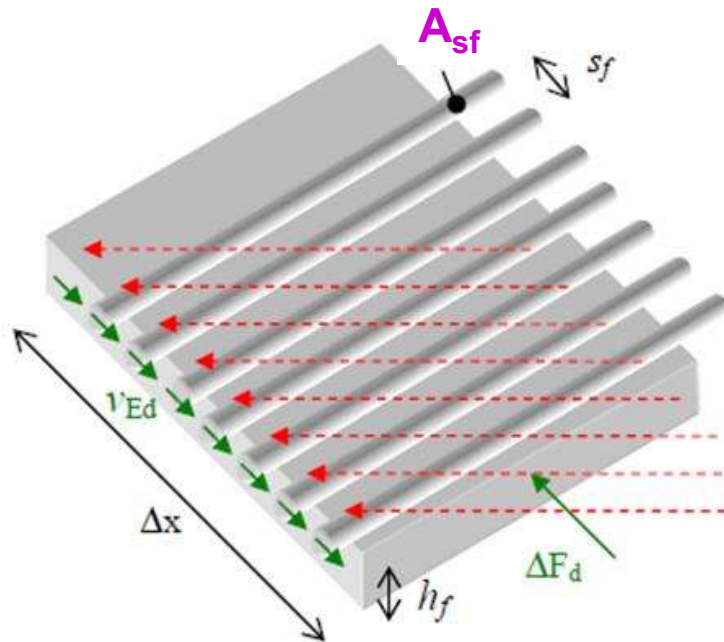


$$\theta_f \geq \frac{1}{2} \sin^{-1} \left(\frac{2 v_{Ed}}{v f_{cd}} \right)$$

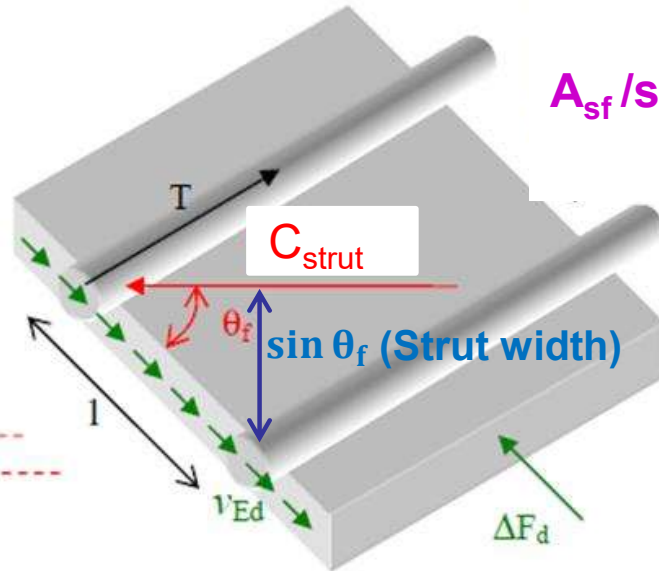
$26.5 \leq \theta_f \leq 45^\circ$ (compressive flange) and $38.6 \leq \theta_f \leq 45^\circ$ (tensile flange)

6 – Flange and web connection

Transverse reinforcement area A_{sf}

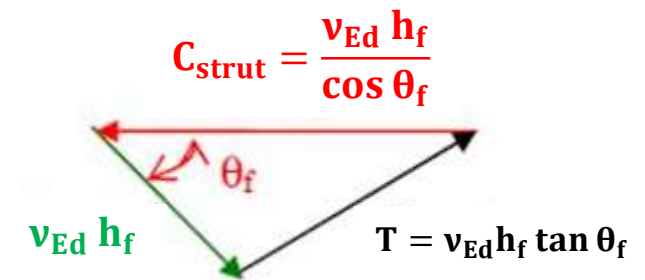


Flange web junction



Unit mesh model

C_{strut} = compressive force within strut
 A_{sf} = transverse reinforcement area
 T = tension force in A_{sf}
 v_{Ed} = longitudinal shear stress



Equilibrium of forces

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Tension force in the transverse reinforcement A_{sf} (unit mesh model): $T = v_{Ed} h_f \tan \theta_f$

Transverse reinforcement per unit length

$$\frac{A_{sf}}{s_f} = \frac{v_{Ed} h_f \tan \theta_f}{f_{yd}}$$

6 – Flange and web connection

Required transverse reinforcement area A_{sf}

If $v_{Ed} \leq k f_{ctd}$ → No extra reinforcement required above those for flexure of the slab

If $v_{Ed} > k f_{ctd}$, three possibilities:

$$\rightarrow \frac{A_{sf}}{s_f} \geq 2 A_{slab} \rightarrow \text{Total section considered} = \frac{A_{sf}}{s_f}$$

$$\rightarrow \frac{A_{sf}}{s_f} < 2 A_{slab} \rightarrow \text{Total section considered} = \frac{A_{sf}}{2 s_f} + A_{slab}$$

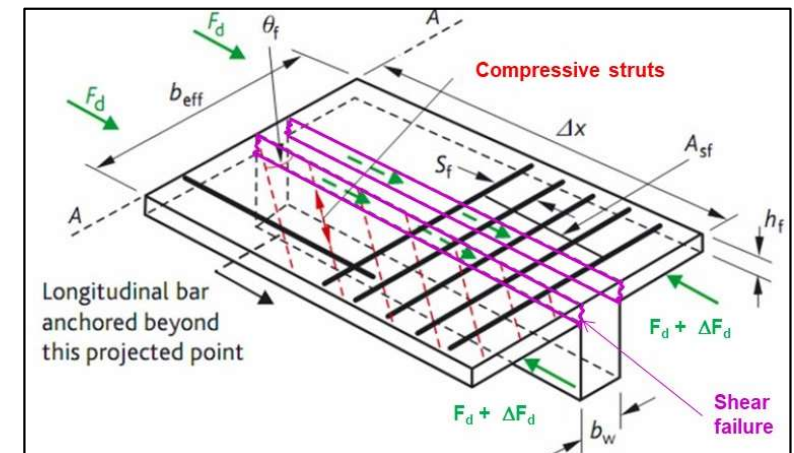
where $A_{slab} = A_{slab,sup} + A_{slab,inf}$ (section of top and bottom flexural reinforcement of the slab in cm^2/m)

→ Reduce the effective flange width b_{eff}

Where k depends on the vertical interface between flange and web

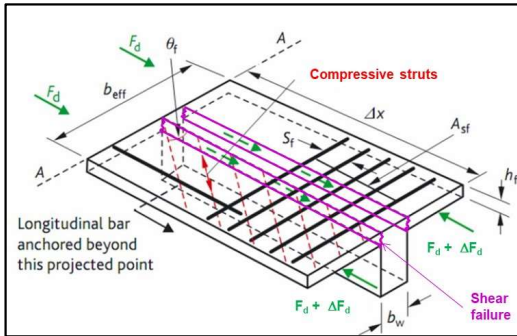
$k = 0.5$ (concrete cast at different times)

$k = 1$ (concrete cast at the same time)



6 – Flange and web connection

$$b_{eff}, b_w, h, d, d', A_s, A'_s, A_{slab} V_{Ed}$$



Variation of the normal force ΔF_d in the flange over Δx

$$\Delta F_d = \frac{\Delta M_{Ed}}{z} \frac{b_{eff} - b_w}{2 b_{eff}}$$

Longitudinal shear stress v_{Ed}

$$v_{Ed} = \frac{\Delta F_d}{h_f \Delta x}$$

No extra reinforcement
needed

YES

$$v_{Ed} \leq k f_{ctd}$$

NO

Inclination angle of the strut θ_f

$$\theta_f \geq \frac{1}{2} \sin^{-1} \left(\frac{2 v_{Ed}}{v f_{cd}} \right)$$

$$26.5 \leq \theta_f \leq 45^\circ \text{ (compressive flange) and } 38.6 \leq \theta_f \leq 45^\circ \text{ (tensile flange)}$$

Transverse reinforcement per unit length

$$\frac{A_{sf}}{s_f} = \frac{v_{Ed} h_f \tan \theta_f}{f_{yd}}$$

Total transverse reinforcement area

$$\frac{A_{sf}}{2 s_f} + A_{slab}$$

YES

$$\frac{A_{sf}}{s_f} < 2 A_{slab}$$

NO

Total transverse reinforcement area

$$\frac{A_{sf}}{s_f}$$

Symbols

Majuscules latines

- A Action accidentelle
- A Aire de la section droite
- A_c Aire de la section droite du béton
- A_p Aire de la section de l'armature ou des armatures de précontrainte
- A_s Aire de la section des armatures de béton armé
- $A_{s,min}$ Aire de la section minimale d'armatures
- A_{sw} Aire de la section des armatures d'effort tranchant
- D Diamètre du mandrin de cintrage
- D_{Ed} Endommagement total dû à la fatigue
- E Effet des actions
- $E_c, E_{c(28)}$ Module d'élasticité tangent à l'origine ($\sigma_c = 0$) pour un béton de masse volumique normale à 28 jours
- $E_{c,eff}$ Module d'élasticité effectif du béton
- E_{cd} Valeur de calcul du module d'élasticité du béton
- E_{cm} Module d'élasticité sécant du béton
- $E_c(t)$ Module d'élasticité tangent à l'origine ($\sigma_c = 0$) au temps t pour un béton de masse volumique normale
- E_p Valeur de calcul du module d'élasticité de l'acier de précontrainte
- E_s Valeur de calcul du module d'élasticité de l'acier de béton armé

Symbols

- EI Rigidité en flexion
- EQU Equilibre statique
- F Action
- F_d Valeur de calcul d'une action
- F_k Valeur caractéristique d'une action
- G_k Valeur caractéristique d'une action permanente
- I Moment d'inertie de la section de béton
- L Longueur
- M Moment fléchissant
- M_{Ed} Valeur de calcul du moment fléchissant agissant
- N Effort normal
- N_{Ed} Valeur de calcul de l'effort normal agissant (traction ou compression)
- P Force de précontrainte
- P_0 Force initiale à l'extrémité active de l'armature de précontrainte immédiatement après la mise en tension
- Q_k Valeur caractéristique d'une action variable
- Q_{fat} Valeur caractéristique de la charge de fatigue
- R Résistance
- S Efforts et moments internes (solllicitations)
- S Moment statique
- SLS Etat-limite de service (ELS)
- T Moment de torsion
- T_{Ed} Valeur de calcul du moment de torsion agissant
- ULS Etat-limite ultime (ELU)
- V Effort tranchant
- V_{Ed} Valeur de calcul de l'effort tranchant agissant

Symbols

Minuscules latines

- a Distance
- a Donnée géométrique
- Δa Tolérance pour les données géométriques
- b Largeur totale d'une section droite ou largeur réelle de la table d'une poutre en T ou en L
- b_w Largeur de l'âme des poutres en T, en I ou en L
- d Diamètre ; Profondeur
- d Hauteur utile d'une section droite
- d_g Dimension nominale supérieure du plus gros granulat
- e Excentricité
- f_c Résistance en compression du béton
- f_{cd} Valeur de calcul de la résistance en compression du béton
- f_{ck} Résistance caractéristique en compression du béton, mesurée sur cylindre à 28 jours
- f_{cm} Valeur moyenne de la résistance en compression du béton, mesurée sur cylindre
- f_{ctk} Résistance caractéristique en traction directe du béton
- f_{ctm} Valeur moyenne de la résistance en traction directe du béton
- f_p Résistance en traction de l'acier de précontrainte
- f_{pk} Résistance caractéristique en traction de l'acier de précontrainte
- $f_{p0,1}$ Limite d'élasticité conventionnelle à 0,1 % de l'acier de précontrainte
- $f_{p0,1k}$ Valeur caractéristique de la limite d'élasticité conventionnelle à 0,1 % de l'acier de précontrainte
- $f_{0,2k}$ Valeur caractéristique de la limite d'élasticité conventionnelle à 0,2 % de l'acier de béton armé
- f_t Résistance en traction de l'acier de béton armé
- f_{tk} Résistance caractéristique en traction de l'acier de béton armé
- f_y Limite d'élasticité de l'acier de béton armé
- f_{yd} Limite d'élasticité de calcul de l'acier de béton armé

Symbols

- f_{yk} Limite caractéristique d'élasticité de l'acier de béton armé
- f_{ywd} Limite d'élasticité de calcul des armatures d'effort tranchant
- h Hauteur
- h Hauteur totale de la section droite
- i Rayon de giration
- k Coefficient ; Facteur
- l (ou l ou L) Longueur ; Portée
- m Masse
- r Rayon
- $1/r$ Courbure en une section donnée
- t Epaisseur
- t Instant considéré
- t_0 Age du béton au moment du chargement
- u Périmètre de la section droite de béton dont l'aire est A_c
- u, v, w Composantes du déplacement d'un point
- x Profondeur de l'axe neutre
- x, y, z Coordonnées
- z Bras de levier des forces internes

Symbols

Minuscules grecques

- α Angle ; Rapport
- β Angle ; Rapport ; Coefficient
- γ Coefficient partiel
- γ_A Coefficient partiel relatif aux actions accidentelles A
- γ_C Coefficient partiel relatif au béton
- γ_F Coefficient partiel relatif aux actions F
- $\gamma_{C,fat}$ Coefficient partiel relatif à la fatigue du béton
- $\gamma_{F,fat}$ Coefficient partiel relatif aux actions de fatigue
- γ_G Coefficient partiel relatif aux actions permanentes G
- γ_M Coefficient partiel relatif à une propriété d'un matériau, tenant compte des incertitudes sur la propriété elle-même, sur les imperfections géométriques et sur le modèle de calcul utilisé
- γ_P Coefficient partiel relatif aux actions associées à la précontrainte P
- γ_Q Coefficient partiel relatif aux actions variables Q
- γ_S Coefficient partiel relatif à l'acier de béton armé ou de précontrainte
- $\gamma_{S,fat}$ Coefficient partiel relatif à l'acier de béton armé ou de précontrainte sous chargement de fatigue
- γ_f Coefficient partiel relatif aux actions, compte non tenu des incertitudes de modèle
- γ_g Coefficient partiel relatif aux actions permanentes, compte non tenu des incertitudes de modèle
- γ_m Coefficient partiel relatif à une propriété d'un matériau, seules les incertitudes sur la propriété du matériau étant prises en compte
- δ Incrément / coefficient de redistribution
- ζ Coefficient de réduction / coefficient de distribution
- ε_c Déformation relative en compression du béton
- ε_{c1} Déformation relative en compression du béton au pic de contrainte f_c
- ε_{cu} Déformation relative ultime du béton en compression
- ε_u Déformation relative de l'acier de béton armé ou de précontrainte sous charge maximale
- ε_{uk} Valeur caractéristique de la déformation relative de l'acier de béton armé ou de précontrainte sous charge maximale
- θ Angle
- λ Coefficient d'élancement
- μ Coefficient de frottement entre les armatures de précontrainte et leurs conduits
- ν Coefficient de Poisson

Symbols

- ν Coefficient de réduction de la résistance du béton fissuré en cisaillement
- ξ Rapport de la capacité d'adhérence des armatures de précontrainte à la capacité d'adhérence des armatures de béton armé
- ρ Masse volumique du béton séché en étuve, en kg/m^3
- ρ_{1000} Valeur de la perte par relaxation (en %), 1 000 heures après la mise en tension, à une température moyenne de 20°C
- ρ_l Pourcentage d'armatures longitudinales
- ρ_w Pourcentage d'armatures d'effort tranchant
- σ_c Contrainte de compression dans le béton
- σ_{cp} Contrainte de compression dans le béton due à un effort normal ou à la précontrainte
- σ_{cu} Contrainte de compression dans le béton correspondant à la déformation ultime en compression ϵ_{cu}
- τ Contrainte tangente de torsion
- ϕ Diamètre d'une barre d'armature ou d'une gaine de précontrainte
- ϕ_n Diamètre équivalent d'un paquet de barres
- $\phi(t, t_0)$ Coefficient de fluage, définissant le fluage entre les temps t et t_0 , par rapport à la déformation élastique à 28 jours
- $\phi(\infty, t_0)$ Valeur finale du coefficient de fluage
- ψ Coefficients définissant les valeurs représentatives des actions variables
 - ψ_0 pour les valeurs de combinaison
 - ψ_1 pour les valeurs fréquentes
 - ψ_2 pour les valeurs quasi-permanentes

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